

**A PROJECT WORK REPORT ON
ANALYSIS OF HEAT TRANSFER ENHANCEMENT AND
PRESSURE DROP IN PLATE HEAT EXCHANGERS USING
NANOFLUIDS**

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

MECHANICAL ENGINEERING



Submitted by

Abhishek Kashyap (210710002005)

Aroty Tokbipi (210710002014)

Chandra Sekhar Rai (210710002028)

Dhruba Jyoti Das (210710002033)

Under the supervision of

Dr. Bijay Kumar Roy

(Associate Professor, Department of Mechanical Engineering)

JORHAT ENGINEERING COLLEGE

JORHAT

ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

2025

CANDIDATES' DECLARATION

We hereby declare that the work which is being presented in this report entitled "Analysis of heat transfer enhancement and pressure drop in plate heat exchangers using nanofluid" is an authentic record of our own work carried out during the period from August 2024 to December 2024 under the supervision of Dr. Bijay Kumar Roy of Mechanical Engineering Department, Jorhat Engineering College, Jorhat, 785009 (Assam).

The matter presented in this report has not been submitted for the award of any other degree of this or any other university.

Abhishek Kashyap (210710002005)

Aroty Tokbipi (210710002014)

Chandra Sekhar Rai (210710002028)

Dhruba Jyoti Das (210710002033)

SUPERVISOR'S CERTIFICATE

This is certified that the above statement made by the candidates is true to the best of my knowledge.

Date:

Dr. Bijay Kumar Roy

Place: Jorhat, Assam

(Associate Professor)

CERTIFICATE OF EXAMINATION

The viva-voce examination of the above candidates of 8th semester B.Tech Mechanical Engineering on their project has been held on 09/05/2025 and found satisfactory.

Supervisor

H.O.D.

External Examiner

CANDIDATES' DECLARATION ON PLAGIARISM

We hereby declare that the content of the work which is being presented in this report entitled “Analysis of heat transfer enhancement and pressure drop in plate heat exchangers using nanofluids” has been checked by <https://paperpal.com/tools/plagiarism-checker>, and the plagiarism index is found to be 9 %.

Signature of the Supervisor

Dr. Bijay Kumar Roy
(Associate Professor)

Signatures of the Candidates

Abhishek Kashyap
(210710002005)

Aroty Tokbipi
(210710002014)

Chandra Sekhar Rai
(210710002028)

Dhruba Jyoti Das
(210710002033)

Date:

Place: Jorhat, Assam

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ABSTRACT

This project investigates heat transfer enhancement and pressure drop in plate heat exchangers using nanofluids, with a focus on improving their thermal performance while addressing flow resistance challenges. In industrial use the plate heat exchangers with chevron plate design is used where pressure drop management is an issue. This study focuses on altering the plate design to cylindrical extrusion.

Repeated flow simulations are performed with various plate with chevron design and cylindrical extrusion design with different extrusion height, keeping other flow and design parameters fixed, to study the heat transfer and pressure drop results. The results of flat plate, chevron plate and cylindrical extrusion plates are compared and further simulations for cylindrical extrusion plate with different extrusion height are performed.

The results are validated and the effect of the extrusion height is established using statistical tests like anova. Further the results are plotted and fitted to obtain a general equation for heat transfer rate as well as pressure drop in terms of height of the cylindrical extrusions. The general equations are used to optimize the heat exchanger and find an optimum point of extrusion height for which there is a balanced trade off between heat transfer rate and pressure drop.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

Symbols	Description
T_{h1}	Temperature of hot fluid at inlet (°C)
T_{h2}	Temperature of hot fluid at outlet (°C)
T_{c1}	Temperature of cold fluid at inlet (°C)
T_{c2}	Temperature of cold fluid at outlet (°C)
U	Overall Heat Transfer Coefficient (W/m ² .K)
A	Total Heat Transfer Area (mm ²)
C_{min}	The minimum heat capacity rate (J/s.K)
ε	Effectiveness
ρ_{nf}	Density of nanofluid (kg/m ³)
ρ_p	Density of nano particles (kg/m ³)
ρ_b	Density of base fluid (kg/m ³)
Φ	Nanoparticle Concentration in base fluid
μ_{nf}	Dynamic viscosity of nanofluid (m Pa.s)
K_{nf}	Thermal conductivity of nanofluid (W/m.K)
C_{Pnf}	Specific Heat of nanofluid (J/kg.K)
C_{Pp}	Specific Heat of nano particles (J/kg.K)
C_{Pb}	Specific Heat of base fluid (J/kg.K)
D_p	Diameter of the port (mm)
L_v	Vertical distance between centre of ports (mm)
L_h	Horizontal distance between ports (mm)
L_w	Plate width inside gasket (mm)
L_p	Plate length (mm)
β	Chevron angle
b	Mean Channel Spacing (mm)
P	Plate pitch (mm)
P_c	Corrugation pitch (mm)
t	Plate thickness (mm)
A_p	Plate effective heat transfer area (mm ²)
F	Plate area enlargement factor
D_e	Hydraulic diameter (mm)

m_{cf}	Mass flow rate of cold fluid (kg/s)
m_{hf}	Mass flow rate of hot fluid (kg/s)
V_{cf}	Volume flow rate of cold fluid (m ³ /s)
V_{hf}	Volume flow rate of hot fluid (m ³ /s)
Q	Heat transfer rate (kW)
P_1	Pressure at inlet (kPa)
P_2	Pressure at exit (kPa)
ΔP	Pressure drop (kPa)
P_{c1}	Inlet pressure of cold fluid (kPa)
P_{h2}	Inlet pressure of hot fluid (kPa)
c	Number of groups
n	Sample size
α	Significance level
n_{ex}	Number of extrusions
D_{ex}	Diameter of extrusion
h_{ex}	Height of extrusion

CHAPTER 1: INTRODUCTION

1.1 Heat Exchangers

Heat exchanger is a device used to transfer thermal energy between two or more fluids, or a solid surface and a fluid, or particulate solid and a fluid. Heat exchangers have a variety of applications, design and wide range of size and capacity.

1.2 Classification of Heat Exchangers

A. According to the transfer process:

- I. Indirect contact type: Does not allow direct contact or mixing of the hot and cold fluids.
- II. Direct contact type: Hot and cold fluids comes in direct contact with each other.

B. According to geometry:

- I. Recuperators: Transfers heat directly from one fluid to another. Examples: Shell and tube HE, Plate HE
- II. Regenerators: Has an intermediate storage tank. Examples: Fixed bed regenerators, rotating wheel regenerators

C. According to the construction:

- I. Tubular: Shell and tube type, spiral tube, double pipe etc
- II. Plate Heat Exchangers: Flat plate, Spiral, Plate coiled, Printed Circuit etc
- III. Extended surface type: Plate fin, tube fin

D. According to relative direction of fluid flow:

- I. Parallel flow type: Both hot and cold fluid flows in the same direction. Examples: Oil coolers, Water heaters etc.
- II. Counter flow type: The hot and cold fluid flows in opposite direction.
- III. Cross flow type: The two fluids cross each other at right angles.

E. According to the physical state of the fluid:

- I. Condenser: The condensing fluid remains at constant temperature while the cold fluid gains temperature.

II. Evaporator: The cold fluid remains at constant temperature and the temperature of the hot fluid changes.

1.3 Applications of Heat Exchangers

1. HVAC Systems: Heat exchangers are used to transfer heat between indoor and outdoor environments in heating, ventilation, and air conditioning (HVAC) systems, improving energy efficiency and comfort.
2. Refrigeration and Air Conditioning: In these systems, heat exchangers manage heat absorption and rejection, enabling cooling cycles essential for appliances like refrigerators and air conditioners.
3. Power Plants: Heat exchangers are critical in power plants for transferring heat between fluids, such as in steam turbines, where steam is condensed back into water.
4. Petrochemical and Chemical Industries: They aid in temperature regulation during chemical reactions and processing, often cooling or heating chemical mixtures to optimize reaction rates.
5. Automotive Industry: Vehicle radiators and oil coolers use heat exchangers to dissipate excess heat from engines, maintaining optimal operating temperatures.
6. Renewable Energy: In solar thermal systems and geothermal plants, heat exchangers transfer heat from collected energy into usable thermal or electrical energy forms.
7. Food and Beverage Processing: They help in pasteurization, sterilization, and heating processes, preserving food quality while managing temperatures effectively.
8. Waste Heat Recovery: Heat exchangers reclaim waste heat from industrial processes, re-purposing it to reduce energy consumption and costs.

1.4 Analysis of Heat Exchangers

1.4.1 The LMTD method

The Logarithmic Mean Temperature Difference (LMTD) method is a widely used approach in heat exchanger analysis, particularly for designing and evaluating the performance of heat exchangers. It is applicable to both parallel-flow and counter flow heat exchangers, as well as shell-and-tube types with multiple passes. The LMTD method calculates the heat transfer in a heat exchanger by averaging the temperature

difference between the hot and cold fluids across the exchanger length. This average is not an arithmetic mean; instead, it is a logarithmic mean because the temperature difference between the two fluids varies along the length.

The LMTD is given by:

$$\text{LMTD} = (T_{H2} - T_{C2}) - (T_{H1} - T_{C1}) / \ln \{(T_{H2}-T_{C2})/(T_{H1}-T_{C1})\}$$

The rate of heat transfer Q is given by:

$$Q = UA (\text{LMTD})$$

1.4.2 The NTU method

The Number of Transfer Units (NTU) method is an alternative approach to the Logarithmic Mean Temperature Difference (LMTD) method for analyzing heat exchangers. It's especially useful when inlet temperatures and heat exchanger characteristics (like flow configuration and surface area) are known, but outlet temperatures are unknown.

The formula for NTU is given by:

$$\text{NTU} = UA/C_{\min}$$

Where, $C_{\min}$ = The minimum heat capacity rate out of the two fluids = $\min(C_{\text{hot}}, C_{\text{cold}})$

Effectiveness of a heat exchanger:

The NTU method is based on the effectiveness of the heat exchanger, which is the ratio of the actual heat transfer to the maximum possible heat transfer. The maximum possible heat transfer occurs when the fluid with the lower heat capacity rate undergoes the maximum possible temperature change.

$$\varepsilon = \text{Actual heat transfer/Maximum possible heat transfer}$$

1.5 Plate Heat Exchangers

Plate heat exchangers (PHEs) are a type of heat exchanger used to transfer heat between fluids efficiently. They consist of a series of thin, corrugated plates, typically

made of stainless steel or other metals, stacked together. These plates create alternating channels for hot and cold fluids to pass through, maximizing the surface area available for heat transfer and enabling efficient energy exchange between the fluids. The unique design of plate heat exchangers allows for a compact structure, making them ideal for applications where space is limited. Each plate has a chevron or herringbone pattern, which increases turbulence as the fluid flows over it. This turbulence improves heat transfer rates, making plate heat exchangers highly efficient compared to other types, like shell and tube heat exchangers. However, the increased turbulence can also lead to a higher pressure drop across the exchanger, which may require balancing for optimal performance.

1.5.1 Advantages of plate heat exchangers

Plate heat exchangers offer several advantages, making them popular in various applications:

1. **High Efficiency:** The large surface area provided by the corrugated plates allows for efficient heat transfer, often more effective than other types of heat exchangers like shell and tube exchangers.
2. **Compact Design:** PHEs are space-saving due to their compact structure, making them ideal for installations where space is limited.
3. **Easy Maintenance and Cleaning:** The plates in PHEs can be disassembled, cleaned, and reassembled relatively easily, which is especially beneficial in industries requiring strict hygiene standards, like food and pharmaceuticals.
4. **Flexible Capacity:** Plate heat exchangers can be easily expanded or modified by adding or removing plates, allowing for capacity adjustments based on demand changes.
5. **Low Fluid Volume Requirement:** PHEs require a smaller amount of fluid to operate, which reduces the energy needed for heating or cooling and can lead to faster response times.

6. High Turbulence, Low Fouling: The corrugated plate design creates turbulence at lower flow rates, reducing the risk of fouling by preventing sediment build-up on the plates.

7. Cost-Effectiveness: Their modular design and efficient heat transfer make PHEs cost-effective over their life cycle, with relatively lower maintenance costs.

9. Suitability for Multiple Fluids: PHEs can handle a wide range of fluids, including those with high viscosity or particle content, and they are adaptable to different thermal requirements and fluid types.

1.5.2 Construction of a Plate Heat Exchanger

A plate heat exchanger (PHE) is constructed with a series of thin, corrugated metal plates, usually made from stainless steel or titanium. These plates are stacked closely together, and each plate has a chevron or herringbone pattern to increase surface area and create turbulence, enhancing heat transfer. Each plate has gaskets along its edges, which seal the channels and direct the hot and cold fluids through alternate gaps between plates. The plates are clamped together in a frame, with two end plates at the front and back. Fluids enter and exit through ports on the frame, flowing in a counter flow or parallel flow arrangement. This modular design allows easy modification by adding or removing plates to adjust capacity, and the clamped structure enables easy disassembly for cleaning and maintenance.

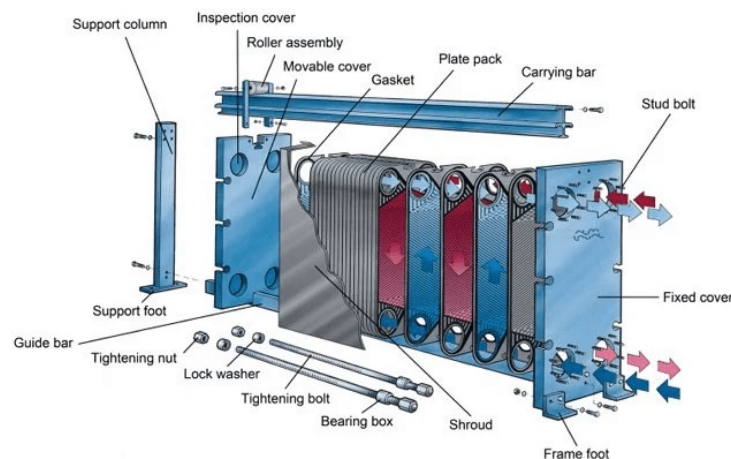


Figure 1.1 Construction of a Plate Heat Exchanger

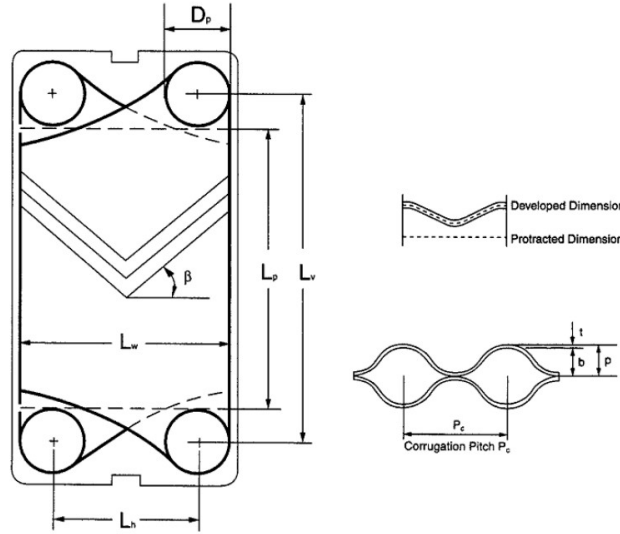


Figure 1.2 Plate of a Plate Heat Exchanger

1.5.3 Parameters of the plate

The total effective heat transfer area of the plate is given by,

$$A_p = F L_w L_p$$

Where, F is called plate area enlargement factor (varies from 1.15 to 1.25), and it is the ratio between the plate effective heat transfer area, A_p and the designed area (product of length and width $L_p \times L_w$)

The plate length and width can be calculated as,

$$L_p = L_v - D_p$$

$$L_w = L_h + D_p$$

The hydraulic diameter of the channel is given by,

$$D_e = 2bF$$

The heat transfer rate is given by,

$$Q = UA_p (\text{LMTD})$$

1.6 Nanofluids as Working Fluid in Heat Exchangers

Nanofluids are advanced heat transfer fluids that consist of a base fluid, such as water, ethylene glycol, or oil, with nanoparticles suspended within it. These nanoparticles, typically made from materials like metal oxides (e.g., Al_2O_3 , CuO), metals (e.g., Cu , Ag), or carbon-based materials (e.g., graphene, carbon nanotubes), have sizes ranging

from 1 to 100 nm. The inclusion of nanoparticles significantly enhances the thermal conductivity, specific heat, and overall heat transfer performance of the base fluid.

1.6.1 Desirable properties of nanofluids

1. High Thermal Conductivity: Nanoparticles increase the thermal conductivity of the base fluid, improving its ability to transfer heat.
2. Enhanced Convection Heat Transfer: Nanofluids exhibit improved convective heat transfer coefficients compared to conventional fluids.
3. Stability: Proper dispersion techniques ensure the nanoparticles remain evenly distributed, preventing sedimentation.
4. Tailored Properties: By selecting specific nanoparticles and concentrations, the thermal properties of nanofluids can be customized for various applications.

1.6.2 Reasons for heat transfer enhancement

1. Higher Thermal Conductivity: Nanoparticles, such as metals and metal oxides, possess much higher thermal conductivity compared to conventional base fluids like water or oil. When dispersed in a base fluid, they significantly increase the fluid's ability to conduct heat, thereby improving overall heat transfer rates.
2. Increased Surface Area: The small size and large quantity of nanoparticles result in a much larger surface area for heat exchange compared to conventional fluids. This increased surface area facilitates greater interaction between the fluid and the heat transfer surfaces, enhancing thermal performance.
3. Brownian Motion of Nanoparticles: The random motion of nanoparticles within the fluid, known as Brownian motion, contributes to additional energy exchange between particles and the surrounding fluid. This microscopic agitation improves the thermal conductivity of the nanofluid.
4. Improved Convection: Nanofluids exhibit higher convective heat transfer coefficients because the suspended nanoparticles modify the fluid's flow behavior,

creating turbulence at the microscopic level. This enhanced turbulence boosts the convective heat transfer process in heat exchangers.

5. Thermophoresis Effect: Under temperature gradients, nanoparticles tend to migrate from hotter regions to cooler regions. This thermophoretic movement redistributes heat more effectively within the fluid, improving overall heat transfer performance.

6. Reduced Boundary Layer Thickness: The presence of nanoparticles disrupts the thermal boundary layer near heat transfer surfaces, reducing its thickness. A thinner boundary layer means less thermal resistance, allowing more efficient heat transfer from the surface to the fluid or vice versa.

7. Enhanced Fluid Thermal Capacity: Certain nanofluids increase the effective specific heat capacity of the fluid, enabling it to store and transfer more heat. This is especially useful in applications where fluids must absorb large amounts of heat energy.

8. Improved Flow Behavior: Nanofluids can exhibit non-Newtonian flow properties, where their viscosity changes under varying flow conditions. This adaptive behavior can optimize heat transfer in systems with varying flow dynamics, ensuring better performance under specific operating conditions

1.6.3 Disadvantages of using nanofluids

1. Stability Issues: Nanoparticles tend to aggregate and settle over time, especially if the dispersion is not stable. This can lead to uneven thermal properties and reduced heat transfer performance.

2. High Cost: Producing and maintaining nanofluids can be expensive due to the cost of nanoparticles and the need for advanced preparation techniques to ensure stability.

3. Increased Pressure Drop: The addition of nanoparticles can increase the viscosity of the fluid, leading to higher frictional losses and increased pressure drop in heat exchangers. This may require more energy for pumping, reducing overall system efficiency.

4. Erosion and Corrosion: Nanoparticles can cause erosion or abrasion of heat exchanger surfaces, particularly at high concentrations or velocities. Additionally, certain nanoparticles may react chemically with the surfaces, leading to corrosion.
5. Clogging of Channels: In micro- or mini-channel heat exchangers, the accumulation of nanoparticles can lead to clogging, reducing the efficiency and potentially damaging the system.
6. Health and Safety Concerns: The potential toxicity of nanoparticles to humans and the environment is a significant concern. Improper handling or disposal of nanofluids may pose risks.

CHAPTER 2: LITERATURE REVIEW

2.1 Literary Survey

Various research has been conducted studying the affect of use of nanofluid in heat transfer applications, particularly in plate heat exchangers. The effect of various parameters of the design and flow conditions are also studied.

Yousefi, T. et al (2012) conducted an experimental investigation into the effect of Al₂O₃–H₂O nanofluid on the efficiency of flat-plate solar collectors. The study analyzed thermal performance with Al₂O₃ nanoparticle concentrations of 0.1%, 0.2%, 0.3%, and 0.4% by volume at a constant flow rate of 1 liter per minute. Findings demonstrated that increasing Al₂O₃ concentration improved the heat transfer rate, enhancing the collector's efficiency by up to 15% at a concentration of 0.4% compared to using water alone. Temperature differentials between the inlet and outlet showed an increase in thermal gain, with nanofluid use raising the average temperature by 3–6 °C. Additionally, the heat transfer coefficient of the Al₂O₃–H₂O nanofluid was notably higher than that of conventional fluids, supporting better thermal absorption and transfer in the collector. These results suggest that Al₂O₃–H₂O nanofluids can significantly improve the thermal efficiency of solar collectors, potentially aiding in the advancement of sustainable energy technologies.

Stalin, P. M. et al. (2019) conducted an energy, economic, and environmental analysis of a flat-plate solar collector using CeO₂/water nanofluid in a forced circulation system of 100 LPD capacity. The investigation examined flow rates (1–3 L/min) and CeO₂ concentrations (0.01–0.1%). Results showed a 28.07% efficiency increase compared to water-only systems, with the highest efficiency at 0.05% concentration and 2 L/min. Exergy analysis revealed optimal performance at 0.05%, with minimized entropy generation and enhanced Bejan number. Additionally, a 24.52% reduction in collector area was achieved, saving 300.2 MJ of energy and offsetting 175 kg of CO₂. The CeO₂/water nanofluid system also showed a reduced payback period of 2.12 years compared to conventional systems, highlighting its economic viability and environmental benefits.

Ekramian, E. et al (2014) performed a numerical investigation on the heat transfer performance of nanofluids in flat plate solar collectors. Using computational fluid dynamics (CFD) simulations, they analyzed three nanofluids—MWCNT/water, Al₂O₃/water, and CuO/water—at mass concentrations of 1%, 2%, and 3%. The study found that CuO/water nanofluid exhibited the highest thermal efficiency and heat transfer coefficient, surpassing both Al₂O₃/water and MWCNT/water. A comparison of simulation results with experimental data showed close alignment, validating the numerical approach's reliability. The research underscores nanofluids' potential to enhance solar collector efficiency by improving heat transfer characteristics.

Mostafizur et al. (2023) conducted a thermodynamic analysis of a flat-plate solar collector using various CuO-MWCNT hybrid nanofluids in five different base fluids: water, ethylene glycol, methanol, radiator coolant, and engine oil. The study examined nanoparticle concentrations from 1% to 3% and flow rates between 1 and 3.5 L/min, focusing on factors like entropy generation, exergy efficiency, and pressure drop. CuO-MWCNT/water nanofluid showed the highest exergy efficiency and lowest entropy generation across all volume fractions, while methanol-based nanofluid provided the best heat transfer but demanded less pumping power. Results suggest CuO-MWCNT/water as a promising fluid for solar collectors to maximize efficiency and minimize energy loss.

Koli and Shah (2022) provided an analysis on the design considerations for plate heat exchangers (PHEs), focusing on optimizing thermal performance and efficiency. The authors outlined the critical design parameters, including plate material, thickness, chevron angle, and plate arrangement, which impact both heat transfer and pressure drop. Their study examined the effect of varying chevron angles—specifically, angles between 30 and 60 degrees on flow turbulence and heat transfer rates. They presented data indicating that increasing the chevron angle from 30 to 45 degrees enhances heat transfer by approximately 20%, but also raises the pressure drop by around 15%. The paper also discussed the influence of plate spacing on performance, showing that closer spacing improves heat transfer but can lead to higher pumping requirements. Additionally, the authors evaluated material choices, with stainless steel and titanium demonstrating favorable thermal conductivity and corrosion resistance. The study

concluded by recommending optimal design combinations, suggesting that selecting appropriate chevron angles and plate materials based on operational needs can significantly improve PHE efficiency. The authors called for further studies using different fluids and angles to achieve more refined optimization of PHEs for industrial applications.

Musediq et al. (2021) conducted a computational analysis on the thermal performance of a water-based copper oxide (CuO) nanofluid in a double-pipe, small-scale heat exchanger. The study examined the influence of CuO nanoparticle concentration (ranging from 0.5% to 2% by volume) on heat transfer efficiency and pressure drop. Simulation results showed that increasing the nanoparticle concentration improved thermal conductivity and enhanced heat transfer. For example, a 1.5% CuO concentration resulted in a 25% increase in heat transfer rate compared to pure water. However, the study also noted that this enhancement is accompanied by a significant rise in pressure drop, particularly at higher concentrations, with up to a 30% increase in pumping power requirements. The authors investigated the effects of different Reynolds numbers, finding that heat transfer improvement is more pronounced in turbulent flow regimes. Optimal performance is observed with a CuO concentration of around 1%, balancing heat transfer gains with manageable pressure drops. The study concluded that while CuO nanofluids offer promising enhancements in heat transfer for small-scale heat exchangers, careful consideration of nanoparticle concentration and flow conditions is essential to avoid excessive energy costs.

Brinkman (1952) explored the viscosity characteristics of concentrated suspensions and solutions, presenting a model to estimate viscosity in systems with high particle concentrations. The study derived an expression that accounts for the interactions among suspended particles, particularly in cases where the concentration affects flow resistance. Brinkman's formula extended Einstein's viscosity equation to higher particle volume fractions, proposing that the effective viscosity increases as particle concentration rises. Experimental data demonstrated that viscosity grows exponentially when the volume fraction of suspended particles exceeds approximately 5%, with a notable rise in systems with non-spherical particles. This finding has significant implications for nanofluid applications in heat exchangers, as increased particle concentration, while beneficial for thermal conductivity, may adversely affect

flow characteristics due to the higher viscosity. Brinkman's work remains foundational in understanding the trade-off between heat transfer and fluid flow resistance in systems employing high-concentration suspensions, such as nanofluids used in thermal systems.

Mintsa et al. (2009) present experimentally derived data on the temperature-dependent thermal conductivity of water-based nanofluids, focusing on mixtures containing nanoparticles such as Al_2O_3 and CuO . The study demonstrated that the thermal conductivity of these nanofluids increases with both temperature and nanoparticle concentration, with Al_2O_3 and CuO nanofluids exhibiting enhancements of up to 20% and 24%, respectively, at elevated temperatures around 60°C and a 2% volume concentration. Results indicated that the thermal conductivity of CuO nanofluids shows a slightly more pronounced increase with temperature compared to Al_2O_3 . The authors attributed this behavior to intensified Brownian motion at higher temperatures, which enhances energy transport within the fluid. They also highlighted that at higher nanoparticle concentrations, thermal conductivity improvements are more significant, albeit accompanied by an increase in viscosity. This temperature-dependent data provides crucial insights for the design of heat exchangers, suggesting that nanofluids like CuO can enhance heat transfer effectively in systems operating at elevated temperatures, though with an associated trade-off in flow resistance due to viscosity.

Zhou and Ni (2008) investigated the specific heat capacity of water-based Al_2O_3 nanofluids, providing empirical measurements to assess how nanoparticle concentration affects this property. The study revealed that the specific heat capacity of Al_2O_3 nanofluids decreases with increasing nanoparticle concentration, contrasting with the thermal conductivity enhancement typically observed. For example, at a 2% volume concentration of Al_2O_3 , the specific heat capacity of the nanofluid is approximately 10% lower than that of pure water. This reduction is attributed to the lower specific heat of Al_2O_3 particles relative to water, resulting in an overall decrease in the fluid's ability to store thermal energy. The authors noted that while adding Al_2O_3 nanoparticles improves heat transfer due to increased thermal conductivity, the lower specific heat may impact the nanofluid's effectiveness in applications requiring significant thermal energy absorption. These findings are

significant for heat exchanger design, as they suggest that while Al₂O₃ nanofluids enhance heat transfer, their reduced specific heat could limit performance in applications where heat capacity is critical.

Pak and Cho (1998) conducted an experimental study on the hydrodynamic and heat transfer properties of fluids containing submicron metallic oxide particles, focusing on Al₂O₃ and TiO₂ nanofluids. Their research examined the effects of particle concentration and Reynolds number on heat transfer enhancement and fluid flow behavior. Results indicated that adding nanoparticles significantly improves the heat transfer coefficient, with Al₂O₃ nanofluids showing up to a 12% increase and TiO₂ nanofluids achieving around 8% enhancement at a 3% volume concentration. However, this enhancement comes with an increase in viscosity, leading to higher pressure drops. The study reported that for particle concentrations above 1%, the pressure drop increases by approximately 20-30%, impacting pumping power requirements. At higher Reynolds numbers, the heat transfer improvement is more noticeable, especially in the turbulent regime. Pak and Cho concluded that while metallic oxide nanofluids offer considerable heat transfer benefits, the rise in viscosity and associated pressure drops require careful consideration for efficient design in heat exchanger applications. Their work highlights the balance between thermal performance and flow resistance when employing nanofluids.

Unverdi and Islamoglu (2016) investigated the heat transfer and pressure drop characteristics in a chevron-type plate heat exchanger using Al₂O₃/water nanofluids. The study explores various concentrations of Al₂O₃ nanoparticles (0.5% to 2% by volume) and examines their impact on thermal performance and flow resistance within the heat exchanger. Results showed that increasing nanoparticle concentration enhances the heat transfer rate, with a 1.5% concentration of Al₂O₃ leading to a 22% improvement compared to pure water. However, this increase in thermal efficiency is accompanied by a significant rise in pressure drop, particularly at higher concentrations, where pressure drop increased by 25-35%. The research also analyzed the influence of chevron angles on heat transfer, finding that angles of 35 to 45 degrees improve the heat transfer rate by up to 18% but at a cost of elevated pressure drops by approximately 15-20%. The authors concluded that Al₂O₃ nanofluids offer substantial heat transfer benefits in chevron-type PHEs, especially in applications

requiring high thermal efficiency, though attention must be given to managing pressure drop to ensure optimal performance.

Wu and Zhao (2013) reviewed the advancements in nanofluid heat transfer and critical heat flux (CHF) enhancement, analyzing current research and identifying gaps hindering broader engineering applications. The review highlighted that nanofluids, particularly those with metallic nanoparticles like Al_2O_3 , CuO , and TiO_2 , significantly improve thermal conductivity and CHF under specific conditions. For instance, at a 1% volume concentration, Al_2O_3 nanofluids demonstrated up to a 15% increase in thermal conductivity and a 30% CHF enhancement, providing potential benefits for cooling systems in high-heat applications such as nuclear reactors. However, the authors note challenges related to stability, viscosity increases, and material compatibility that impact practical usage. Data show that increased viscosity, particularly at higher nanoparticle concentrations, results in substantial pressure drop penalties, with some cases reporting up to a 40% increase in pumping requirements. The paper identifies a need for comprehensive studies addressing long-term stability, nanoparticle dispersion, and the effects of surface roughness in real-world conditions. Wu and Zhao concluded that bridging these research gaps is crucial for transitioning nanofluids from laboratory research to viable solutions in engineering applications, particularly in high-performance thermal management systems.

Shanbedi et al. (2021) investigated the effects of different surfactants, including SDS, on the stability and thermophysical properties of CNTs in water. The study focuses on optimizing the heat transfer performance of multi-walled carbon nanotubes (MWCNTs)-water nanofluids in a plate heat exchanger. Past research has extensively explored nanofluids as heat transfer enhancers due to their superior thermal properties. Key studies have examined the use of various nanoparticles, including metal and metal oxides, in improving the thermal conductivity and convective heat transfer coefficients in heat exchangers. Specifically, prior work has shown that carbon nanotubes (CNTs) exhibit high thermal conductivity compared to other nanoparticles, making them promising candidates for enhancing the performance of heat exchangers. Their results indicate that CNT-based nanofluids enhance heat transfer coefficients significantly when optimized for parameters like concentration, surfactant type, and temperature. Similarly, experimental work by Sunder et al. on magnetic nanofluids in

tube heat exchangers found improved convective heat transfer performance, emphasizing the role of particle concentration and Reynolds number in determining heat transfer rates.

Sözen, Khanlari, and Çiftçi (2019) investigated the performance of a plate heat exchanger using TiO₂-water nanofluid as the working fluid. Nanofluids, due to their enhanced thermal properties, have been shown to improve heat transfer efficiency in heat exchangers. Recent studies have emphasized using nanoparticles like Al₂O₃, SiO₂, and CeO₂ in water-based fluids to enhance heat transfer in plate heat exchangers, with varying results depending on particle type and concentration. For instance, Tiwari et al. demonstrated that CeO₂-water nanofluid provided better thermal performance than other oxide-based nanofluids, and Kumar et al. found that the performance was influenced by plate geometry, specifically chevron angle. In the current study, Sözen et al. conducted experiments and numerical simulations with TiO₂ nanofluids, measuring performance at different flow rates and temperatures. Results showed that nanofluids improved heat transfer coefficients significantly compared to water alone, with the highest enhancement observed at higher flow rates. Additionally, pressure drop was noted as an essential factor, which increased with the nanofluid's use but was not extensively analyzed in terms of optimization.

Gupta et al.(2020) provided insights into the application of nanofluids in plate heat exchangers (PHEs) to enhance thermal performance. Nanofluids, which are dispersions of nanoparticles in base fluids, improve the thermal conductivity of working fluids, resulting in better heat transfer rates in heat exchangers. Carbon nanotubes (CNTs), particularly multi-walled carbon nanotubes (MWCNTs), are shown to be highly effective due to their superior thermal properties. Studies revealed that nanofluids, especially those with MWCNTs and hybrid combinations (e.g., Al₂O₃/MWCNT), significantly improve heat transfer performance while potentially reducing pressure drop and pumping power requirements. The application of hybrid nanofluids in PHEs has also been investigated, with findings indicating that Al₂O₃/MWCNT mixtures achieve high stability and heat transfer rates with minimal additional pressure drop. Several studies included in this review highlighted the benefits of using nanofluids over conventional fluids in heat exchangers, noting improvements in convective heat transfer coefficients under both laminar and

turbulent flow conditions. Experiments with varying nanoparticle types, sizes, and concentrations have provided data supporting the use of CNT-based nanofluids for achieving optimal performance, particularly under high Reynolds numbers and turbulent conditions. This performance enhancement is attributed to the high surface area-to-volume ratio and superior thermal properties of CNTs.

Javadi et al. (2013) examined the impact of nanofluids on thermophysical properties and heat transfer in plate heat exchangers, finding that nanofluids—especially those with nanoparticles like Al_2O_3 and TiO_2 —significantly enhance thermal conductivity and convective heat transfer. The study revealed that nanofluid properties, including particle type and concentration, influence heat transfer rates and flow behavior within the exchanger. By comparing nanofluids to conventional fluids, the research aligns with findings by Sadeghipour and Saidur, who also reported improved heat transfer performance using nanofluids in various configurations.

Asadi et al. (2014) studies that explore the role of plate corrugation angles and flow distribution in PHEs. Several studies indicate that modifications to chevron angles can significantly affect fluid dynamics, leading to variations in heat transfer rates and pressure losses. Some studies also address the use of nanofluids to further enhance heat transfer in such systems. Despite extensive research on the effects of chevron angles, there is limited understanding of how non-ideal flow conditions (e.g., uneven distribution or complex fluids) influence the thermal performance of corrugated PHEs. Additionally, further experimental studies are needed to optimize these angles under various operational conditions and for different fluid types.

Moradi et al. (2019) investigated the flow and heat transfer characteristics of multi-walled carbon nanotube (MWCNT)–water nanofluids in a double-pipe heat exchanger with porous media. The study examined various concentrations of MWCNTs (ranging from 0.1% to 0.3% by volume) and analyzes their effects on thermal performance and pressure drop. Results show that incorporating MWCNTs significantly enhances heat transfer, with the 0.3% concentration achieving a 35% improvement compared to pure water. The porous media further intensifies the heat transfer rate by increasing surface contact and turbulence within the heat exchanger. However, the study also recorded a notable increase in pressure drop, especially at higher MWCNT concentrations, where

the pressure drop rises by approximately 20-30%. The authors observed that the flow resistance in porous media compounds this pressure drop effect, emphasizing the need to balance nanoparticle concentration and flow conditions for optimal performance. They concluded that while MWCNT-water nanofluids combined with porous media offer substantial heat transfer benefits, careful control of nanoparticle loading and porous media properties is essential to manage pressure drop effectively. Nanofluids, particularly those containing Multi-Walled Carbon Nanotubes (MWCNTs), have been extensively studied for their superior heat transfer properties. Functionalization techniques, such as covalent bonding with functional groups like cysteine (Cys), are commonly employed to improve the dispersion and stability of MWCNTs in aqueous media, enhancing their thermal conductivity. Previous studies have demonstrated that MWCNT-based nanofluids can significantly improve heat transfer in heat exchangers due to the nanoparticles' high surface area and thermal conductivity. However, as the particle concentration increases, it may lead to higher friction factors, resulting in increased pressure drops and pumping power requirements. The use of such nanofluids in systems like corrugated plate heat exchangers has shown promising results, with improved heat transfer coefficients and Nusselt numbers in turbulent flow regimes.

Walvekar et al. (2016) tried to study enhancement of heat transfer with varying the concentration of CNT material with base fluid water. It was done with a concentric tube Heat Exchanger. Carbon Nanotube (CNT) nanofluids have been widely researched for their potential to enhance heat transfer due to the high thermal conductivity and large surface area of CNTs. Studies have shown that CNT-based nanofluids can significantly improve thermal conductivity and convective heat transfer compared to conventional fluids like water. Dispersion stability of CNTs is crucial, as stable suspensions ensure consistent heat transfer performance. Techniques such as the use of surfactants like Gum Arabic (GA) are often applied to maintain stability by preventing particle agglomeration. Previous studies also indicate that heat transfer performance in turbulent flow conditions aligns well with theoretical models, suggesting the strong applicability of CNT nanofluids in various heat exchanger configurations.

Kumar et al. (2016) studied the effect of variable spacing between plates on the thermal performance of plate heat exchangers (PHEs) using nanofluids. The research investigated different spacing configurations and their impact on heat transfer efficiency and pressure drop when using Al₂O₃ and TiO₂ nanofluids. Results indicated that reducing plate spacing improves heat transfer rates, with narrower gaps enhancing fluid turbulence and promoting better heat transfer. For example, a reduction in spacing from 5 mm to 3 mm with Al₂O₃ nanofluid at a 1% volume concentration led to a 28% increase in heat transfer coefficient. However, this improvement is associated with a substantial increase in pressure drop, nearly 35%, due to restricted flow passages. TiO₂ nanofluids demonstrated similar performance trends, with a slightly lower heat transfer enhancement but a comparable pressure drop increase. The study concluded that optimizing plate spacing is critical when using nanofluids, as it can significantly enhance thermal performance while balancing pressure drop. The authors recommended further research into adjustable plate configurations and alternative nanofluids to improve PHE efficiency without compromising flow characteristics excessively.

Pantzali et al. (2009) examined the impact of nanofluids on the performance of a miniature plate heat exchanger with a modulated surface, focusing on thermal efficiency and pressure drop. The study tested various nanofluids, including Al₂O₃ and CuO at concentrations of 1% and 2% by volume, to analyze their effect on heat transfer and flow resistance. Results showed that the modulated surface enhances turbulence, which, combined with the higher thermal conductivity of nanofluids, improves heat transfer rates significantly. For instance, the use of 2% CuO nanofluid increased the heat transfer coefficient by up to 40% compared to pure water, with the modulated surface further boosting performance by an additional 10-15%. However, this enhancement also led to a noticeable increase in pressure drop, with some configurations experiencing a rise of up to 25%. The study concluded that while nanofluids and surface modulation together enhance heat exchanger efficiency, careful selection of nanofluid type and concentration is essential to balance heat transfer gains with manageable pressure losses. The authors suggest that these findings have applications in compact heat exchangers where space and efficiency are critical.

Zhang et al. (2019) provided a comprehensive review of techniques used to enhance heat transfer in plate heat exchangers (PHEs), analyzing methods such as surface modification, flow arrangement changes, and the use of nanofluids. The paper highlights that surface modifications, including increased corrugation and baffles, significantly boost heat transfer rates, with corrugated plates enhancing the heat transfer coefficient by up to 30-40% compared to smooth plates. The addition of nanofluids, like CuO or Al₂O₃, further enhances thermal performance due to their higher thermal conductivity; CuO nanofluids at 1-2% concentration improve heat transfer by around 20-25%. However, these techniques often lead to a higher pressure drop, with surface modifications increasing pressure by 10-20% and nanofluids by an additional 15% at higher concentrations. The review also discussed research gaps, emphasizing the need for studies focused on optimizing these enhancement methods to balance heat transfer improvements and energy costs, particularly in sustainable energy systems. Zhang et al. concluded that combining techniques, such as using nanofluids with optimized plate geometries, holds promise for achieving significant PHE performance gains with careful management of flow resistance.

Abou Elmaaty et al. (2017) reviewed the design and thermal performance of corrugated plate heat exchangers (PHEs), emphasizing the impact of plate corrugation on heat transfer efficiency and pressure drop. The review explores various corrugation patterns, such as sinusoidal, chevron, and trapezoidal shapes, and discusses how each pattern influences fluid turbulence and heat transfer. Findings indicated that chevron-patterned plates typically yield the highest heat transfer improvements, increasing the heat transfer coefficient by 25-40% compared to flat plates due to enhanced flow mixing. However, this improvement also resulted in an increased pressure drop, particularly at chevron angles above 45 degrees, which can lead to a 30% rise in pressure drop, affecting pumping requirements. The authors also discussed material considerations, highlighting that stainless steel and titanium are commonly used due to their corrosion resistance and durability. The review concluded by identifying a need for further studies on optimizing corrugation patterns and materials, particularly for applications in sustainable energy systems, where maximizing thermal performance while minimizing energy costs is crucial. Abou Elmaaty et al. recommend combining optimized corrugation angles with compatible nanofluids to achieve balanced performance gains.

Gut and Pinto (2003) presented a study on the modeling of plate heat exchangers (PHEs) with generalized configurations, developing a mathematical model that can adapt to various PHE designs and flow arrangements. Their model incorporated parameters like plate geometry, fluid properties, and flow configurations (e.g., counterflow, parallel flow), aiming to predict heat transfer performance and pressure drop across different setups. The model is validated against experimental data and demonstrates high accuracy, with heat transfer predictions within 5% of measured values. The authors highlighted that variations in plate corrugation angles and channel spacing notably influence thermal efficiency and pressure drop, with steeper corrugation angles enhancing heat transfer by up to 30% but also increasing pressure drop by approximately 25%. Additionally, the study discussed how the model can be applied to optimize PHE configurations for specific applications, such as improving efficiency in compact systems or balancing performance and energy costs in industrial applications. Gut and Pinto concluded that this flexible modeling approach is beneficial for designing customized PHEs that achieve targeted performance levels while managing energy and cost efficiency.

Yan et al. (1999) investigated the condensation heat transfer and pressure drop characteristics of refrigerant R123a in a plate heat exchanger (PHE). The study focused on how different operating conditions, such as refrigerant mass flux and plate geometry, affect thermal performance and flow resistance. Results indicate that higher mass flux significantly improves the heat transfer coefficient, with a 30% increase observed as mass flux rises from 20 to 40 kg/m²s. However, this also leads to a considerable increase in pressure drop, approximately 40% at higher flux rates, which affects pumping requirements. The study found that smaller chevron angles in the plate design contribute to better condensation efficiency, though they also elevate the pressure drop due to restricted flow channels. Yan et al. concluded that PHEs with optimized plate angles and flow rates can enhance condensation heat transfer in refrigerant applications while balancing the trade-offs in pressure drop, suggesting that these findings are applicable for designing efficient refrigeration and HVAC systems.

Al Zahrani et al. (2021) examined heat transfer enhancement in a novel flat plate heat exchanger (PHE), focusing on the effects of structural modifications to plate geometry.

The study introduced a flat plate design with wavy channels intended to increase turbulence and thermal performance without substantially increasing flow resistance. Experimental results showed that the wavy channels improve the heat transfer coefficient by up to 25% compared to conventional flat plate designs due to enhanced mixing of the working fluid. Despite the improved heat transfer, the pressure drop remains relatively low, with only a 10-15% increase, attributed to the streamlined design of the wavy channels. The authors also analyzed the impact of different wave amplitudes and wavelengths, finding that optimized wave patterns can balance thermal efficiency and pressure drop, making the design suitable for compact and high-efficiency applications. Al Zahrani et al. concluded that this novel configuration could be valuable in applications requiring high thermal performance, such as in renewable energy systems, where maximizing efficiency with minimal energy costs is essential.

Han et al. (2003) investigated the condensation characteristics of brazed plate heat exchangers (PHEs) with varying chevron angles, analyzing how angle adjustments impact heat transfer efficiency and pressure drop during condensation. The study examined chevron angles ranging from 25° to 65°, with findings indicating that steeper angles (55° and above) significantly enhance the heat transfer coefficient by increasing fluid turbulence, yielding up to a 40% improvement compared to lower angles. However, this increase in heat transfer performance is accompanied by a substantial rise in pressure drop, reaching up to 30% higher at steep angles, due to the added flow resistance within narrow channel paths. The study highlighted that lower chevron angles are preferable when minimizing pressure drop is a priority, whereas higher angles are advantageous in applications where heat transfer efficiency is critical. Han et al. concluded that optimizing chevron angles based on specific application requirements allows for effective balancing of thermal performance and pressure drop in brazed PHEs, making them suitable for varied applications such as refrigeration and air conditioning.

Yusuf et al. (2024) analyzed the efficiency of plate-type heat exchangers (PHEs) using nanofluids in the primary cooling system of a TRIGA 2000 nuclear reactor, employing Computational Fluid Dynamics (CFD) modeling to evaluate performance. The study examined Al₂O₃ and CuO nanofluids at 1% and 2% volume concentrations

to assess their impact on heat transfer and flow characteristics. Results indicated that both nanofluids enhance the heat transfer coefficient compared to water alone, with CuO nanofluid at 2% concentration achieving up to a 30% improvement. However, the increase in thermal efficiency comes with a rise in pressure drop, estimated at 20% for Al₂O₃ and 25% for CuO at the highest concentration due to greater flow resistance. The authors highlight that optimized nanofluid concentration is necessary to balance heat transfer gains and pressure drop, particularly in nuclear applications where reliability and efficiency are critical. The study concludes that nanofluids hold significant potential to improve cooling efficiency in nuclear reactor systems, though careful control of concentration and flow conditions is essential for practical applications.

Porgar et al. (2024) provided a comprehensive review on the application of nanofluids in various heat exchangers, including plate heat exchangers (PHEs), analyzing how different nanofluids influence heat transfer performance and flow resistance. The review covered numerous nanoparticle types—such as Al₂O₃, CuO, and TiO₂—and examines their effects at various concentrations, highlighting that metal oxide nanofluids generally improve the thermal conductivity of base fluids by 15-30%, depending on concentration and particle size. In PHEs, these enhancements can lead to a 20-40% increase in heat transfer efficiency compared to conventional fluids, with higher concentrations yielding better performance. However, the authors noted that these benefits come at the cost of increased viscosity, often causing a 15-25% rise in pressure drop, which impacts energy consumption. The review identifies key research gaps, including the need for optimal nanofluid formulations that maximize heat transfer while minimizing pressure penalties. Porgar et al. concluded that while nanofluids offer promising enhancements for heat exchanger efficiency, more studies are needed to refine formulations and configurations suitable for specific industrial applications where energy efficiency is crucial.

Lazarevikj et al. (2018) conducted a CFD analysis on heat transfer and fluid flow in a plate heat exchanger (PHE) using nanofluids, with a focus on understanding how nanofluid properties influence thermal efficiency and flow characteristics. The study examines Al₂O₃ and CuO nanofluids at various volume concentrations to assess their impact on the heat transfer coefficient and pressure drop within the PHE channels.

Results revealed that Al₂O₃ nanofluid at a 2% concentration enhances the heat transfer coefficient by approximately 25%, while CuO nanofluid yields a slightly higher improvement of around 30% under similar conditions. However, these enhancements come with increased pressure drops, estimated at 20% for Al₂O₃ and 25% for CuO, due to the higher viscosity associated with the nanoparticle suspensions. The authors found that while nanofluids improve thermal performance, careful concentration control is necessary to balance efficiency gains with manageable pressure drops. Lazarevikj et al. concluded that CFD modeling is a valuable tool for optimizing nanofluid usage in PHE applications, allowing for precise adjustments to meet specific performance and energy efficiency requirements in various industrial contexts.

Heris et al. (2024) performed a CFD-based performance analysis of hybrid MWCNT-SiO₂ nanofluids in plate heat exchangers (PHEs), aiming to enhance heat transfer with minimal pressure drop. The study explored various concentrations of MWCNT-SiO₂ hybrid nanofluids and compares their performance with single-component nanofluids. Results showed that the hybrid nanofluid at 1.5% concentration achieves a 30% improvement in the heat transfer coefficient over traditional fluids while only increasing the pressure drop by 10%, demonstrating a favorable balance between efficiency and flow resistance. The unique combination of MWCNTs, which boost thermal conductivity, and SiO₂, which helps reduce fluid viscosity, allows for effective heat transfer with lower viscosity-related penalties. The authors concluded that hybrid nanofluids are advantageous in PHE applications where enhanced thermal performance is needed without significant increases in energy costs. This study highlighted the potential of hybrid nanofluids as a sustainable solution for improving PHE efficiency in energy-intensive industries.

Wcislik and Taler (2024) presented an economic and exergy analysis of TiO₂ + SiO₂ hybrid nanofluid in a plate heat exchanger system used for solar installations. The study evaluates the thermal performance of the hybrid nanofluid, composed of titanium dioxide (TiO₂) and silicon dioxide (SiO₂) nanoparticles suspended in ethylene glycol, assessing its efficiency in solar energy systems. The results showed that the hybrid nanofluid improves heat transfer by up to 35% compared to pure

ethylene glycol due to enhanced thermal conductivity from the nanoparticles. In terms of exergy analysis, the system with the hybrid nanofluid exhibits a significant increase in exergy efficiency, indicating that the system is not only more thermally efficient but also operates with reduced irreversibility. Economic analysis revealed that although the use of hybrid nanofluid increases initial costs, the enhanced efficiency leads to energy savings that result in quicker payback periods, making it economically viable for long-term use in solar applications. The authors conclude that the $\text{TiO}_2 + \text{SiO}_2$ hybrid nanofluid offers both thermodynamic and economic benefits for improving the performance of heat exchanger systems in solar energy installations.

Kumar et al. (2015) provided a comprehensive review on the application of nanofluids in plate heat exchangers (PHEs), focusing on the effects of various nanoparticles on heat transfer performance and flow characteristics. The review discussed the enhancement of thermal conductivity in base fluids through the addition of nanoparticles like Al_2O_3 , CuO , TiO_2 , and SiO_2 , and how these nanofluids improve the heat transfer coefficient in PHEs. The study highlighted that nanofluids can increase heat transfer rates by 20-40%, depending on particle concentration and type. However, the increased viscosity and density associated with nanofluids also lead to higher pressure drops, typically 10-30% higher than base fluids, which can affect pump power requirements. The authors emphasized the trade-off between thermal performance and pressure drop, suggesting that the use of optimized nanofluid formulations can help balance these factors. The review also covers challenges such as nanoparticle stability, dispersion, and the need for further experimental and computational studies to optimize nanofluid-based PHE designs. Kumar et al. concluded that nanofluids offer promising potential for enhancing PHE performance, especially in applications requiring compact heat exchangers with high thermal efficiency.

Pandey et al. (2024) analyzed the performance of a plate-type steam condenser using nanofluids, focusing on the enhancement of heat transfer efficiency and overall system performance. The study investigated various nanofluids, including Al_2O_3 and CuO , to examine their impact on the heat transfer coefficient and pressure drop within the condenser system. Results showed that the use of nanofluids increases the heat transfer rate by up to 25% compared to conventional fluids, due to the enhanced

thermal conductivity of the nanofluids. However, this improvement comes with a corresponding increase in pressure drop, particularly at higher concentrations, which is attributed to the increased fluid viscosity. The study also evaluated the effects of temperature and flow rate on nanofluid performance in steam condensers, concluding that nanofluids, especially at optimal concentrations, provide a significant boost in thermal efficiency without excessively increasing the operational cost. Pandey et al. emphasized that careful selection of nanofluid type and concentration is crucial to achieving an optimal balance between thermal performance and pressure drop. The study suggested that nanofluids could be a promising solution for improving the efficiency of plate-type steam condensers in power generation and industrial applications.

Mangrulkar et al. (2016) conducted an experimental investigation into the convective heat transfer enhancement using alumina/water and copper oxide/water nanofluids. The study focused on comparing the heat transfer performance of these two nanofluids at various concentrations and flow conditions. Results show that both alumina and copper oxide nanofluids significantly enhance heat transfer rates compared to pure water, with copper oxide nanofluid showing the greatest improvement. At a 2% concentration, copper oxide nanofluid achieves up to a 40% increase in the heat transfer coefficient, while alumina nanofluid at the same concentration provides a 30% improvement. However, the increased heat transfer is accompanied by a rise in viscosity and pressure drop, with copper oxide nanofluid exhibiting higher flow resistance than alumina. The study concluded that while both nanofluids enhance thermal performance, copper oxide nanofluid offers a higher heat transfer improvement, making it more suitable for applications where maximizing heat transfer is crucial. Mangrulkar et al. emphasized the importance of optimizing nanofluid concentrations and flow conditions to balance thermal performance with manageable pressure drop for practical applications.

Ghalib et al. (2018) presented an experimental investigation of flow and heat transfer characteristics of nanofluids in a double-pipe heat exchanger, examining the effects of using Al_2O_3 and CuO nanofluids at different concentrations. The study assessed heat transfer coefficients, pressure drops, and overall thermal performance under various flow rates. Results indicated that both nanofluids enhance heat transfer rates

significantly, with Al_2O_3 nanofluid achieving a 20% increase and CuO nanofluid a 30% increase over water as the base fluid. However, this performance boost is accompanied by a 15-25% increase in pressure drop due to the higher viscosity and density of the nanofluids. The researchers observed that higher concentrations of nanofluids lead to improved thermal conductivity, though they also result in greater resistance to flow. Ghalib et al. concluded that while nanofluids are effective for heat transfer enhancement, optimal concentration and flow adjustments are necessary to balance heat transfer improvements with pressure drop to ensure system efficiency in double-pipe heat exchanger applications.

Noghrehabadi and Samimi (2012) studied natural convection heat transfer of nanofluids within a square enclosure, focusing on the effects of thermophoresis and Brownian diffusion. Their analysis examined how nanoparticle concentration and thermal gradients influence heat transfer performance in enclosed spaces. Results demonstrated that thermophoresis and Brownian motion enhance nanoparticle distribution, leading to a more uniform temperature gradient and improved heat transfer rates within the enclosure. The study found that increasing nanoparticle concentration up to a certain point enhances the Nusselt number, indicating stronger convective heat transfer. However, beyond optimal concentration levels, the benefits diminish due to increased viscosity, which restricts fluid movement. Noghrehabadi and Samimi concluded that thermophoresis and Brownian diffusion are significant mechanisms for improving natural convection heat transfer in nanofluids and that their optimal utilization depends on balancing nanoparticle concentration and viscosity. This research supported the potential application of nanofluids for enhanced thermal performance in systems reliant on natural convection within confined spaces.

Pak and Cho (1998) explored the hydrodynamic and heat transfer behavior of dispersed fluids containing submicron metallic oxide particles, focusing on their performance in enhancing heat transfer. The study assesses alumina (Al_2O_3) and titanium dioxide (TiO_2) particles in water at various concentrations to evaluate their effects on thermal conductivity, heat transfer coefficient, and flow resistance. Findings indicated that adding metallic oxide particles significantly improves the heat transfer coefficient, with up to a 20% increase observed at higher particle concentrations. However, this enhancement is accompanied by a substantial rise in

viscosity, resulting in increased pressure drop and flow resistance, which could impact pumping costs. Pak and Cho also observed that the heat transfer rate is highly dependent on particle size, concentration, and fluid velocity, with smaller particles and higher velocities yielding better performance. The study concluded that while metallic oxide nanofluids can significantly enhance heat transfer, the increased viscosity requires careful consideration of system design to manage the trade-off between thermal efficiency and flow resistance.

Albadr, Tayal, and Alasadi (2013) investigated heat transfer performance in a heat exchanger using Al_2O_3 nanofluid at various concentrations. The study examined the impact of Al_2O_3 nanoparticle concentration on the heat transfer coefficient, thermal conductivity, and pressure drop within a shell-and-tube heat exchanger. Results revealed that increasing the concentration of Al_2O_3 nanoparticles significantly enhances the heat transfer rate, with the heat transfer coefficient showing an increase of up to 32% at the highest concentration tested. This improvement is attributed to the nanoparticles' ability to increase the thermal conductivity of the base fluid. However, the study also noted a corresponding increase in pressure drop, especially at higher particle concentrations, due to the elevated viscosity and density of the nanofluid. Albadr et al. concluded that Al_2O_3 nanofluid can effectively boost heat transfer in heat exchangers, although optimizing concentration levels is essential to balance heat transfer efficiency and pressure drop. The findings supported the potential for Al_2O_3 nanofluid use in heat exchanger applications where enhanced thermal performance is required. The findings on mineral oil nanofluids (MO-NF) revealed that nanoparticle addition slightly enhances thermal conductivity, with an average increase of 1.56%, reaching up to 3.87% as temperature rises. This temperature-dependent improvement suggested that nanofluids can significantly benefit heat transfer in systems with variable thermal profiles, with peak conductivity observed at 45°C. Additionally, the decrease in viscosity for both MO and MO-NF with temperature rise indicates reduced flow resistance, which is advantageous for managing pressure drop in heat exchangers. Notably, the minimal impact of TiO_2 nanoparticles on viscosity highlighted the potential to optimize thermal performance without substantially increasing flow resistance.

Sözen, Khanlari, and Çiftçi (2019) conducted an experimental and numerical analysis to enhance the thermal performance of a plate heat exchanger using TiO₂ nanofluid as the working fluid. Their study reported that introducing TiO₂ nanoparticles to deionized water markedly improves the heat transfer coefficient, with an increase of up to 23% compared to the base fluid. The study specifically examined the effects of fluid flow rate and inlet temperature on heat transfer, finding that the heat transfer coefficient rises with increasing flow rate. For example, at a flow rate of 0.6 L/min, the heat transfer coefficient for the nanofluid was observed to be around 10% higher than for deionized water alone. As the flow rate reached 1.2 L/min, this enhancement rose to approximately 23%, underscoring the effectiveness of TiO₂ nanofluid in enhancing thermal conductivity in high-flow scenarios. Furthermore, the study supports the use of CFD modeling as a reliable tool for analyzing and predicting the performance of plate heat exchangers, noting that CFD results closely matched experimental data. This dual approach highlighted CFD's applicability in optimizing variables such as temperature, flow rate, and design modifications, making it a valuable method for improving heat exchanger efficiency in similar studies.

Pandya, Shah, Molana, and Tiwari (2020) provided a comprehensive review of heat transfer enhancement in plate heat exchangers using nanofluids. Their study synthesized findings across various nanofluids, including Al₂O₃, CuO, TiO₂, and SiO₂, noting that nanoparticle addition to base fluids enhances thermal conductivity and overall heat transfer coefficients. For instance, TiO₂ and Al₂O₃ nanofluids are highlighted for their capacity to improve heat transfer coefficients by up to 30% in optimized conditions, which is directly influenced by factors like particle concentration, fluid flow rate, and nanoparticle properties. The review discussed that increasing nanoparticle concentration tends to improve heat transfer efficiency, yet can also raise viscosity, which may elevate pumping power requirements. Additionally, the study found that CFD simulations serve as effective tools for predicting heat exchanger performance, supporting optimal parameter adjustments without extensive physical trials. By consolidating research data, the authors underscore nanofluids' potential to enhance heat exchanger design, improve thermal efficiency, and reduce size requirements, affirming CFD's value in facilitating targeted design improvements. This review provided valuable insights by

emphasizing the importance of selecting nanoparticle types and flow rates to achieve balanced heat transfer and pressure drop in plate heat exchangers.

2.2 Summary of literature review

Nanofluids consisting of base fluids with nanoparticle suspensions, such as Al_2O_3 , CuO , TiO_2 , and hybrid nanocomposites, have been promising fluids that significantly improve the performance of heat exchangers, particularly for plate heat exchangers and flat-plate solar collectors. Due to the high thermal conductivity and convective heat transfer coefficients compared to those of the base fluids, nanofluids have been identified to enhance the efficiency of a heat exchanger. Studies indicate that other parameters like chevron angles and flow arrangements, such as counterflow or parallel flow configurations, affect the overall thermal performance, thus leaving a way to optimize systems for certain applications. However, with the advantages of nanofluids come pressure drops challenges. The increased viscosity and density of nanofluids result in higher pressure losses, thus requiring more pumping power. This trade-off between enhanced heat transfer and energy consumption has promoted research into optimizing nanoparticle concentration and flow rates in order to balance system performance. Of the innovations, hybrid nanofluids—such as $\text{TiO}_2+\text{SiO}_2$ or MWCNT-SiO_2 combinations are receiving attention for their capacity to provide superior thermal properties while showing relatively lower pressure drops than single-component nanofluids. Apart from performance metrics, nanofluids also have contributions toward environmental and economic aspects. They help reduce energy consumption and carbon emissions, making them in line with the sustainability goals set by the global community. Their application in renewable energy systems is notable since they provide higher efficiency under variable operating conditions and reduce dependence on conventional energy sources. However, there are practical challenges with nanofluids that need to be resolved to exploit their full potential. Issues such as particle agglomeration, sedimentation, and long-term stability limit their use. Techniques in the preparation of nanoparticles, such as surface functionalization and stabilization agents, are under investigation to overcome these problems. Computational and experimental studies are essential in understanding the behavior of nanofluids. CFD simulations provide insight into flow patterns, temperature distribution, and pressure drop characteristics, while experimental

validations ensure the reliability of simulation results. On the other hand, a real world experimental validation of such materials shows gaps in studies under certain real-world operating conditions. Considering such things, scalability and costs bring additional complexity in putting the nanofluid to large-scale systems. At any cost, the overall finding in recent research puts an implication that nanofluids are transformative for the management of thermal systems. Nanofluids would help revolutionize heat transfer applications in a wide number of industries by removing the technical, practical barriers related to stability, cost, and system integration, thus paving the way to energy-efficient and sustainable technologies.

2.3 Research Gaps

1. Pressure drop management:

While many research highlights the increase in heat transfer rates, especially associated with nanoparticles like Copper Oxide, the pressure drop associated with it is also significant.

2. Impact of plate design:

Studies have been conducted mainly focusing on chevron angle variation while the assessment of non standard patterns are not done. Non standard patterns like plates with extrusions of different shapes and change in the chevron pattern shape are not analyzed.

3. Exergy and Economic Analysis: Few studies consider the cost-effectiveness of nanofluids in renewable energy systems. Future research may examine the lifecycle costs and benefits of nanofluids to facilitate industrial adoption

4. High-Temperature and High-Pressure Applications: The majority of current research focuses on moderate temperature and pressure ranges. Studies targeting extreme operational environments are essential for applications like nuclear reactors or high-efficiency steam condensers, where nanofluid stability and performance could face new challenges.

2.4 Objective of the project

The objective of the project are:

1. To develop a mathematical model of flat plate type, chevron plate type and cylindrical extrusion plate type plate heat exchanger.
2. To perform simulation of the models using MWCNT nanofluid and water as working fluid.
3. To evaluate the heat transfer rate and pressure drop in the heat exchanger by varying extrusion height using Solidworks.
4. To analyse and compare the change in heat transfer and pressure drop for the different plate geometries and optimize the model.

CHAPTER 3: DESIGN OF HEAT EXCHANGER

3.1 Selection of nanofluid

Various studies found that the heat transfer coefficient of nanofluid is slightly more than that of base fluid at same mass flow rate and inlet pressure and increases with the increase in the mass flow rate and volume concentration of the nanofluid. But with increase in volume concentration the viscosity of the nanofluid also increases leading to increased friction factor. Experiments with different nanofluids like CuO, γ -Al₂O₃, TiO₂, MWCNT, SiO₂ all showed the heat transfer rate to increase significantly. SiO₂ showed minimum pressure drop and Al₂O₃ to have highest heat transfer rate. MWCNT nanofluid showed highest heat transfer rate and significantly less pressure drop.

3.1.1 Multi-walled carbon nanotube (MWCNT)

The selection of MWCNT (Multi-Walled Carbon Nanotube) nanofluid for this project was based on its superior thermal and physical properties, which make it more effective than other nanofluids for enhancing heat transfer. MWCNTs have a unique tubular structure with a high aspect ratio, allowing them to conduct heat much more efficiently compared to other nanoparticles. This results in greater heat transfer enhancement, which is critical for improving the performance of the plate heat exchanger. Additionally, MWCNT nanofluids have better dispersion stability, meaning the particles remain evenly distributed in the base fluid for a longer time without settling, ensuring consistent thermal performance. When compared to other nanofluids, MWCNT nanofluids often achieve higher thermal conductivity enhancements with a relatively lower increase in viscosity. This leads to a smaller pressure drop during operation, reducing the energy required for pumping and making the system more energy-efficient. These combined advantages of higher heat transfer enhancement and lower pressure drop compared to other nanofluids make MWCNT nanofluids an excellent choice for this project. By utilizing these properties, the study aims to optimize the thermal performance of the plate heat exchanger and address challenges like managing pressure losses effectively.

3.1.2 Properties of the nanofluid

The density and specific heat of the nanofluid are calculated from empirical relations (3.1) and (3.2) [18] which depends upon the properties of the nanoparticles and the base fluid alone at 20°C, while the dynamic viscosity and thermal conductivity of the nanofluid at 20°C are extrapolated from the data presented in [18]. The nanofluid is used as the cold fluid with 20°C inlet temperature, hence the properties of the nanofluid is calculated at 20°C. Taking concentration of nanoparticle as 1% in base fluid water, the following properties are evaluated.

Table 3.1 Properties of MWCNT particles and water at 20°C

Properties	MWNCT Particles	Water (Base Fluid)
Density (kg/m ³)	2100	998.2
Specific Heat (J/kg.K)	699.475	4181.8

Density of the Nanofluid:

$$\rho_{nf} = \Phi \rho_p + (1-\Phi) \rho_b \text{ -----(3.1)}$$

$$\rho_{nf} = (0.01 \times 2100) + \{(1 - 0.01) \times 998.2\}$$

$$\rho_{nf} = 1009.218 \text{ kg/m}^3$$

Specific Heat of the Nanofluid:

For specific heat of the nanoparticles at 20°C, the following data from [18] are extrapolated.

Table 3.2 Specific heat of MWCNT at different temperatures

Temperature (°C)	Specific Heat (J/kg.K)
25	702
45	711
65	720
85	730

Since the plot is linear, linear interpolation is used to find the value of specific heat at 20°C using matlab. Code for linear interpolation:

```

untitled2.m
/MATLAB Drive/untitled2.m
1 % Given data points
2 Temp = [25, 45, 65, 85]; % Temperature in Celcius
3 Cp = [702, 711, 720, 730]; % Specific heat capacity in J/g.K
4 % Extrapolation point (20°C = 293K)
5 Temp_new = 20; % Temperature in Kelvin
6 % Perform linear extrapolation
7 Cp_new = interp1(Temp, Cp, Temp_new, 'linear', 'extrap');
8 % Display the result
9 fprintf('Extrapolated Specific Heat at %.2f degree celsius is %.3f J/g.K\n', Temp_new, Cp_new);
10 % Plot original data and extrapolated point
11 plot(Temp, Cp, 'bo-', 'LineWidth', 2, 'MarkerSize', 8) % Original data
12 hold on
13 plot(Temp_new, Cp_new, 'rs', 'MarkerSize', 10, 'MarkerFaceColor', 'r') % Extrapolated point
14 xlabel('Temperature (Celcius)')
15 ylabel('Specific Heat Capacity (J/g.K)')
16 title('Linear Extrapolation of Sepsific Heat vs Temperature')
17 grid on
18 legend('Original Data', 'Extrapolated Point', 'Location', 'Best')
19 hold off

```

Command Window

```

>> untitled2
Extrapolated Specific Heat at 20.00 degree celsius is 699.750 J/g.K

```

Figure 3.1 Linear extrapolation of specific heat in matlab

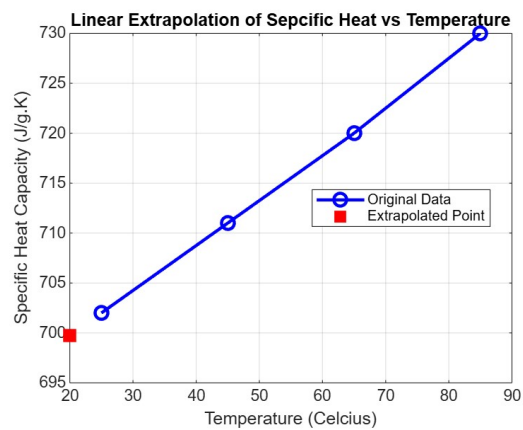


Figure 3.2 Plot of Specific Heat vs Temperature

$$\rho_{nf} \times C_{Pnf} = (1 - \Phi) (\rho_b C_{Pb}) + \Phi (\rho_p C_{Pp}) \text{ -----(3.2)}$$

$$1009.281 \times C_{Pnf} = (0.99 \times 998.2 \times 4181.8) + (0.01 \times 2100 \times 699.75)$$

$$C_{Pnf} = 4109.34 \text{ J/kg.K}$$

Dynamic Viscosity of the Nanofluid:

$$\mu_{nf} = 2.69 \text{ mPa.s}$$

Thermal Conductivity of the Nanofluid:

For thermal conductivity of the nanoparticles at 20°C, the following data from [18] are extrapolated.

Table 3.3 Thermal conductivity of water based MWCNT nanofluid with 1% concentration at different temperatures

Temperature (°C)	Thermal Conductivity (W/m.K)
30	0.69
40	0.712
50	0.742
60	0.766
70	0.789
80	0.801

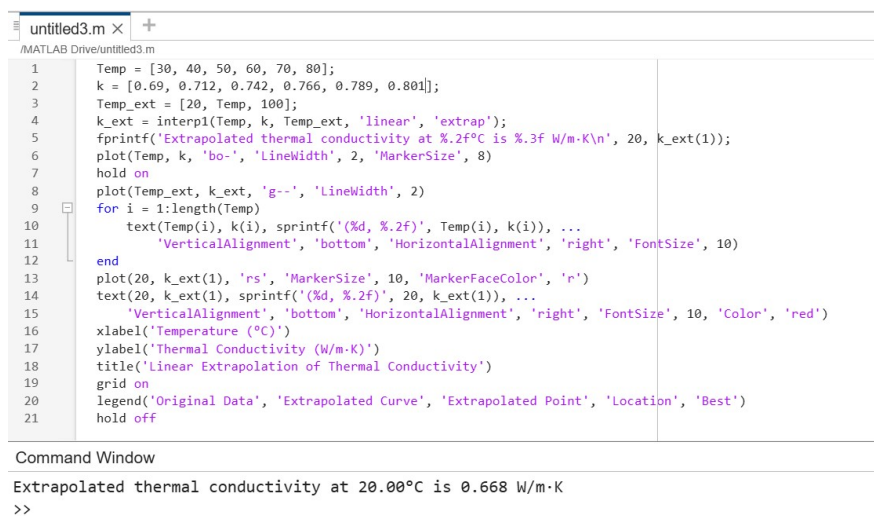


Figure 3.3 Linear extrapolation of thermal conductivity in matlab

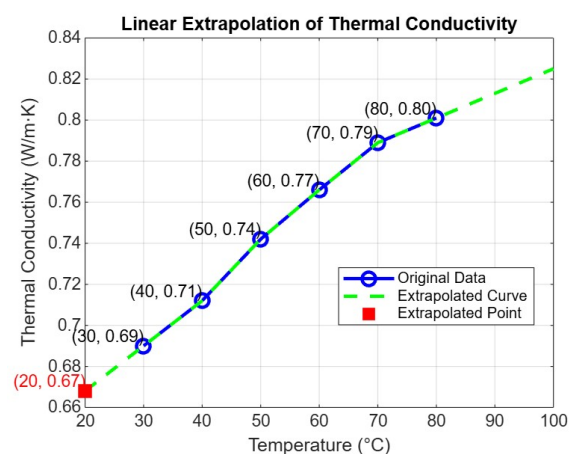


Figure 3.4 Plot of Thermal Conductivity vs Temperature

$$K_{nf} = 0.668 \text{ W/m.K (at } 20^{\circ}\text{C)}$$

Therefore the properties of the water based MWCNT nanofluid (1% concentration) used in the simulation are summarized below.

Table 3.4 Properties of the nanofluid

Properties	Values
Density (kg/m^3)	1009.218
Specific Heat (J/kg.K)	4109.34
Dynamic Viscosity (mPa.s)	2.69
Thermal Conductivity (W/m.K)	0.669

3.2 Components of the Heat Exchanger

3.2.1 Flat plate for reference

The main reference flat plate is design with the following dimensions. Thickness of the plate is taken as standard 1 mm for all the models. The plate material is taken as Copper because of its high thermal conductivity. Various properties of Copper are listed below.

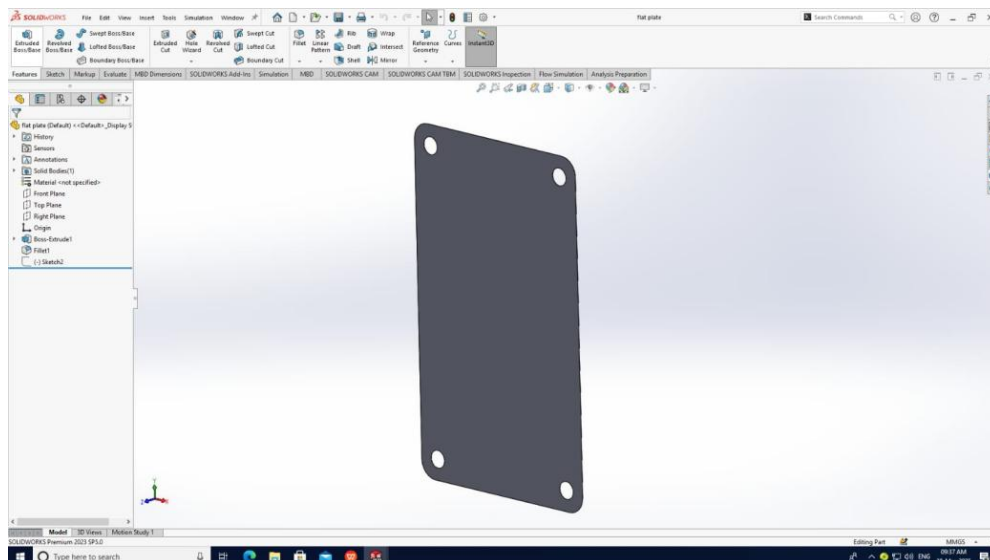


Figure 3.5 Design of the flat plate in Solidworks

Density of Copper = 8960 kg/m^3

Specific heat of Copper:

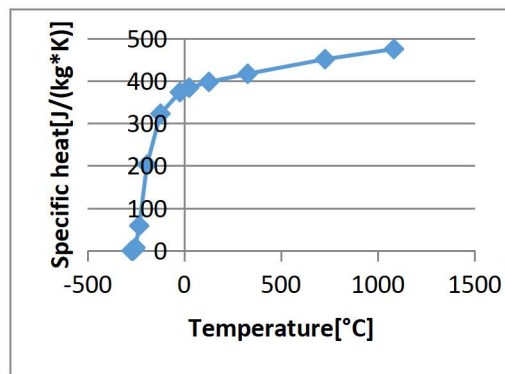


Figure 3.6 Variation of specific heat of copper with temperature

Thermal conductivity of Copper:

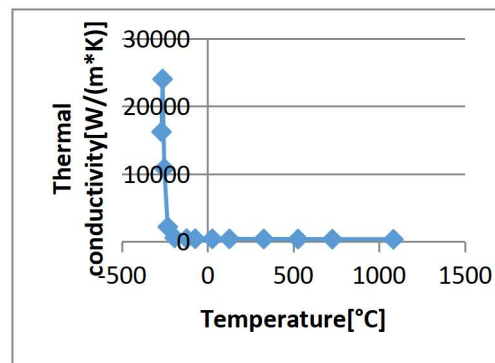
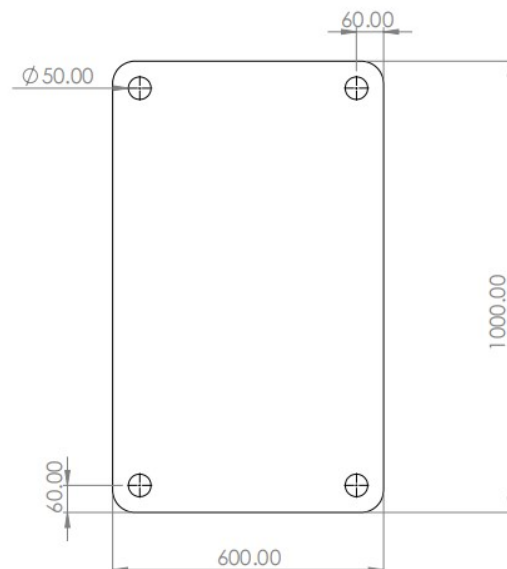


Figure 3.7 Variation of thermal conductivity of copper with temperature

Dimensions of the plate:



All dimensions are in 'mm'

Figure 3.8 Flat Plate

3.2.2 Gasket

The material for the gasket is taken as teflon which has the following properties.

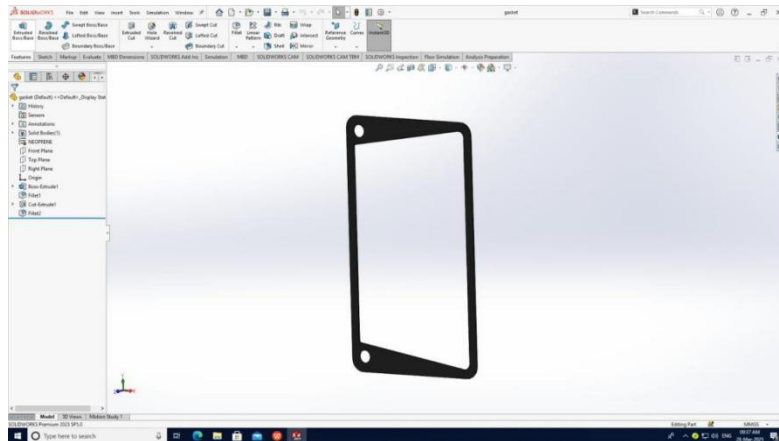


Figure 3.9 Design of the gasket in Solidworks

Density of teflon = 2160 kg/m^3

Specific heat of teflon:

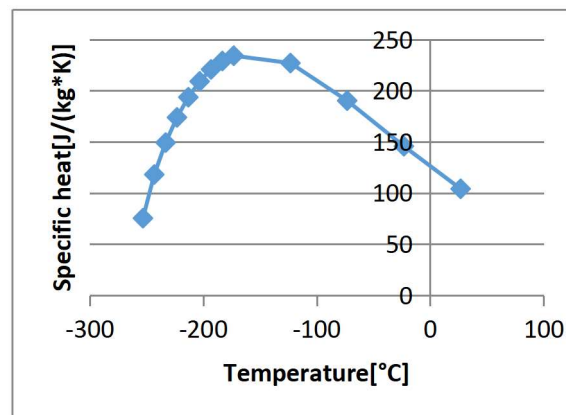


Figure 3.10 Variation of specific heat of teflon with temperature

Thermal conductivity of teflon:

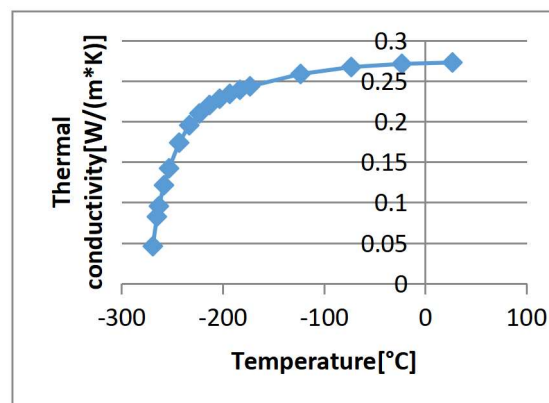
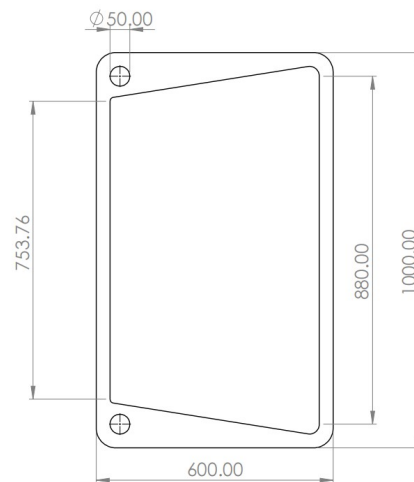


Figure 3.11 Variation of thermal conductivity of teflon with temperature

Dimensions of the gasket:



All dimensions are in 'mm'

Figure 3.12 Gasket

3.2.3 Chevron plates

Chevron plates with rectangular chevron pattern are used with the value of chevron angle to be 30°. Two chevron plates are taken of plate pitch of 0.5 mm and 1mm. The corrugation pitch for both the plates is equal to 20 mm. The material of the chevron plate is same as the flat plate that is Copper.

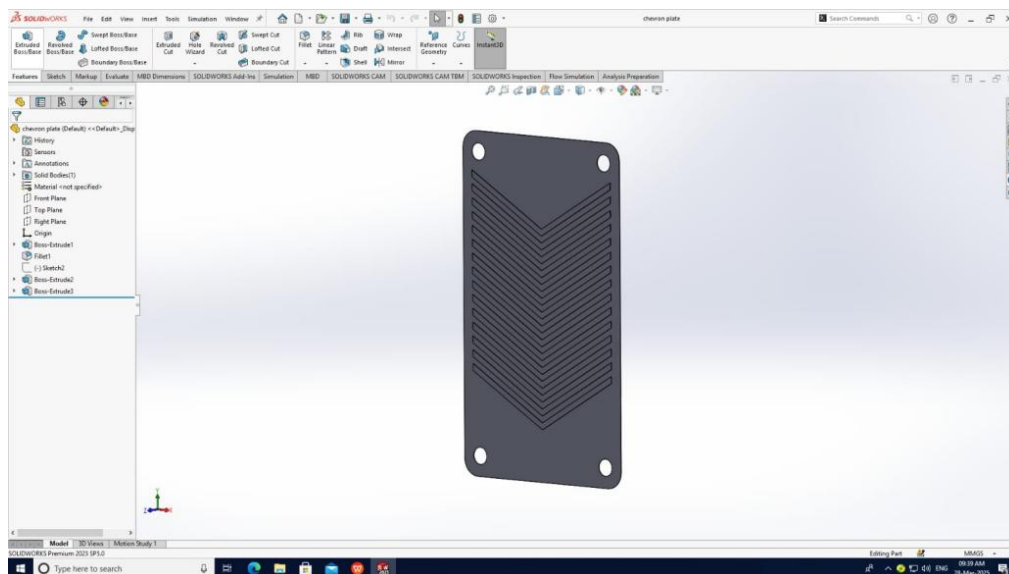
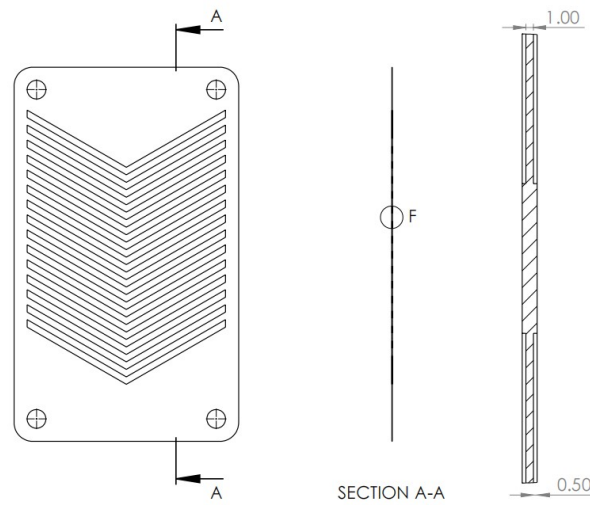


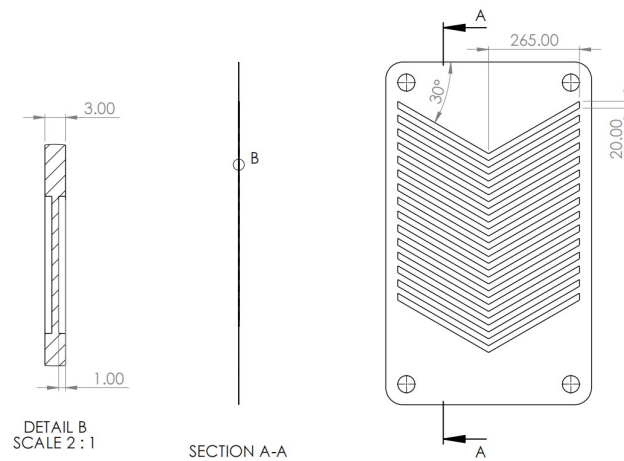
Figure 3.13 Design of the chevron plate in Solidworks

Dimensions:



All dimensions are in 'mm'

Figure 3.14 Chevron plate with plate pitch of 0.5 mm



All dimensions are in 'mm'

Figure 3.15 Chevron plate with plate pitch 1 mm

3.2.4 Plate with cylindrical extrusions

Four different sets of plates having cylindrical extrusions, instead of chevron pattern are designed with a zig-zig pattern of the extrusions with spacing between each line of extrusions to be 20 mm. The diameter of the extrusions are kept as 10 mm. The height of the cylinders are changed in each of the four sets from 0.5 mm, 1 mm, 2 mm, 3mm. The maximum height is limited to 3 mm as the spacing between the two plates (that is the thickness of the gasket) is taken to be 3 mm.

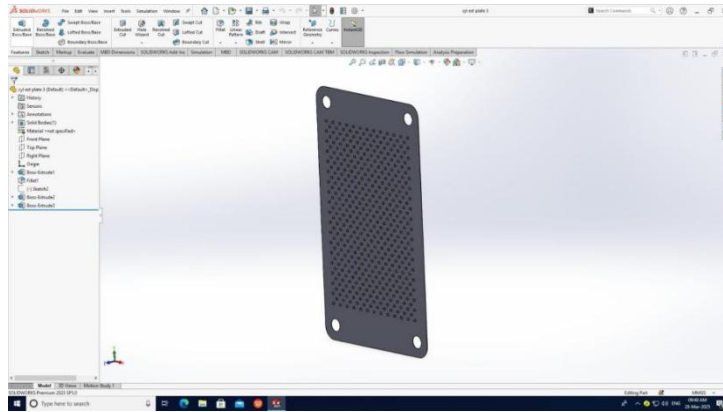
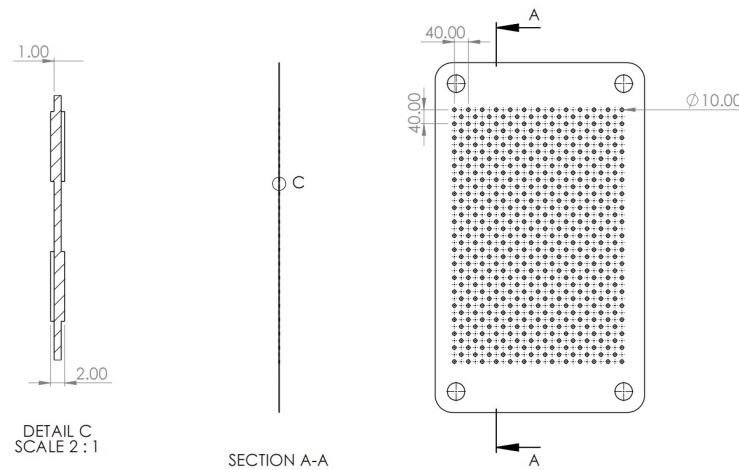


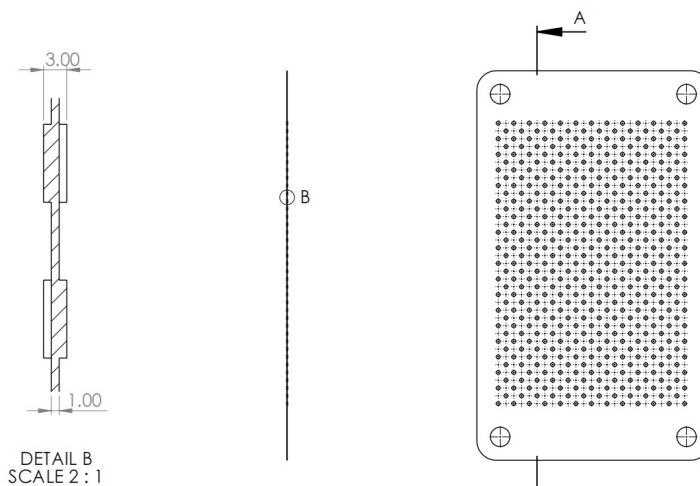
Figure 3.16 Design of the cylindrical extrusion plate in Solidworks

Dimensions:



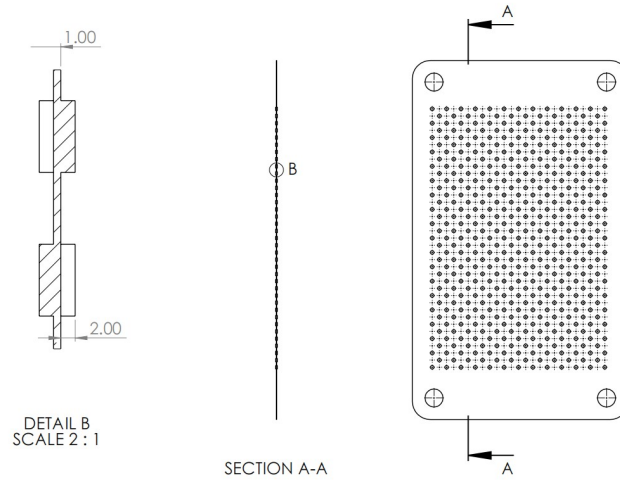
All dimensions are in 'mm'

Figure 3.17 Plate with cylindrical extrusion of 0.5 mm height



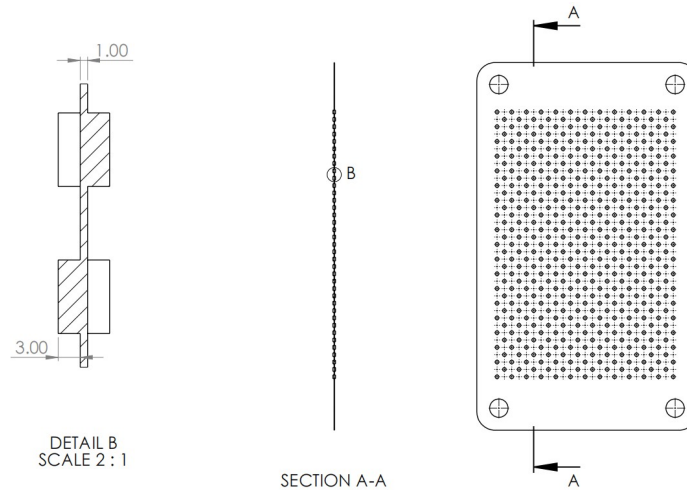
All dimensions are in 'mm'

Figure 3.18 Plate with cylindrical extrusions of 1 mm height



All dimensions are in 'mm'

Figure 3.19 Plate with cylindrical extrusions of 2 mm height



All dimensions are in 'mm'

Figure 3.20 Plate with cylindrical extrusions of 3 mm height

3.2.5 Assembly of the model

The assembly of all the models with the different plates are done. These are one assembly with flat plate, two with chevron plates and four with plate with cylindrical extrusions. Gaskets are sandwiched between the plates to maintain spacing between the plates. The gaskets form the chambers through which the cold and hot fluids alternatively flows in opposite directions. All the assembly of the plate heat exchanger consists of 9 plates and 20 gaskets. The hot and cold fluid flows through 10 chambers each. The width of one chamber is 3 mm which is equal to the width of the gasket.

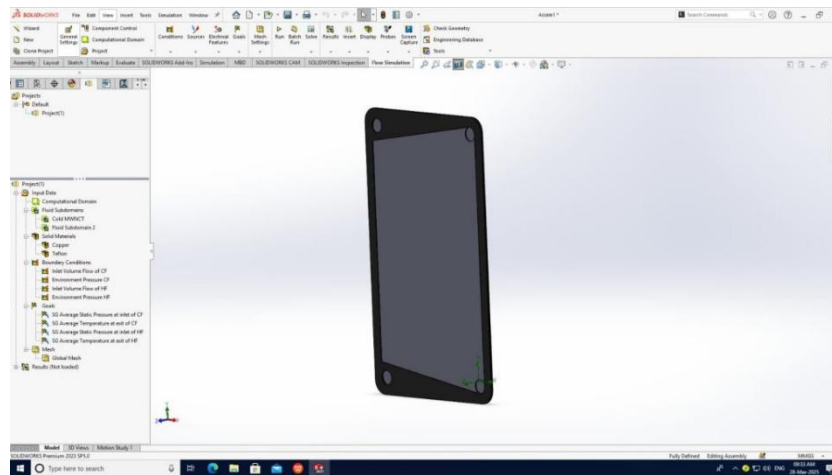
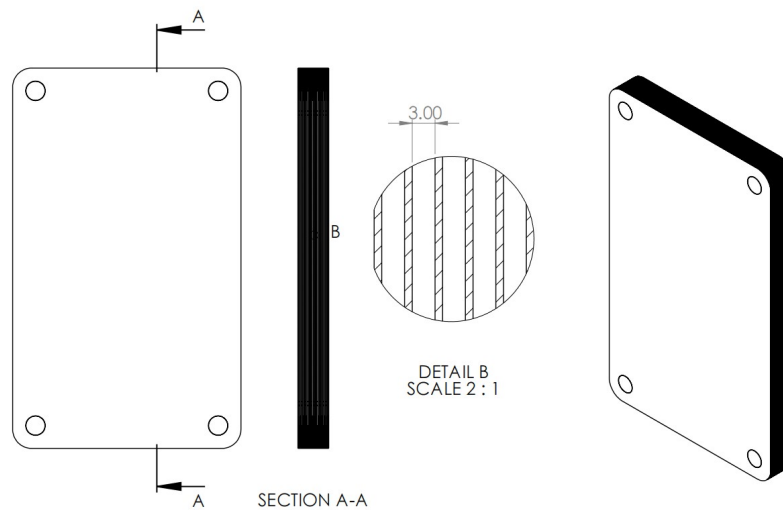


Figure 3.21 Cross sectional view of the flat plate assembly



All dimensions are in 'mm'

Figure 3.22 Assembly of the flat plate heat exchanger

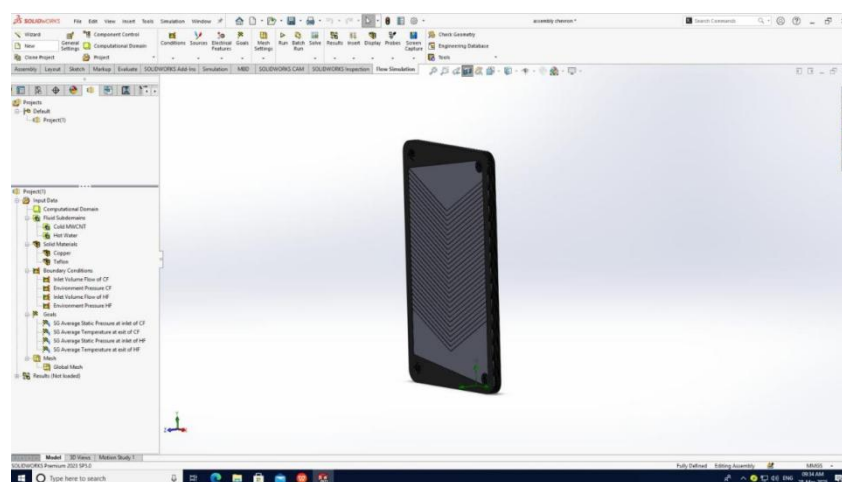
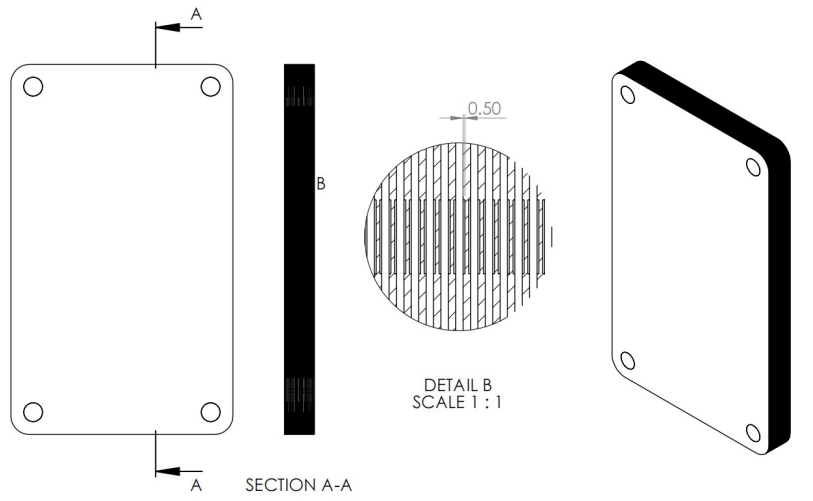
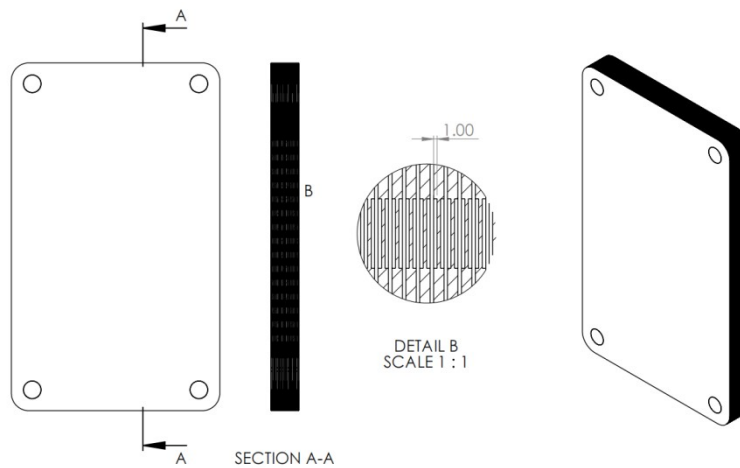


Figure 3.23 Cross sectional view of the chevron plate assembly



All dimensions are in 'mm'

Figure 3.24 Assembly of Chevron plate with 0.5 mm pitch



All dimensions are in 'mm'

Figure 3.25 Assembly of Chevron plate with 1 mm pitch

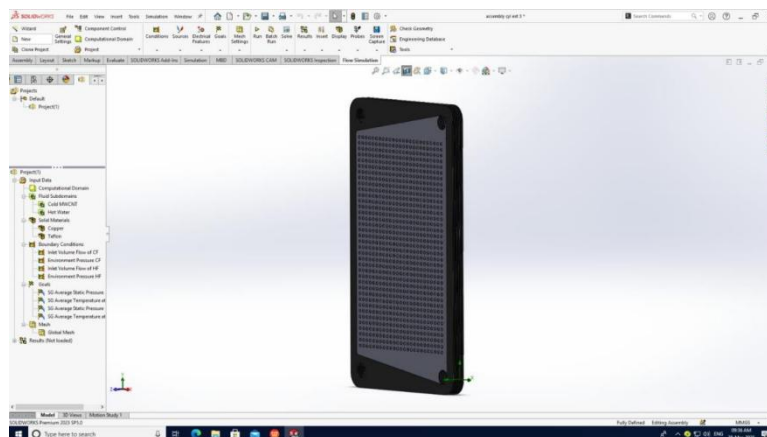
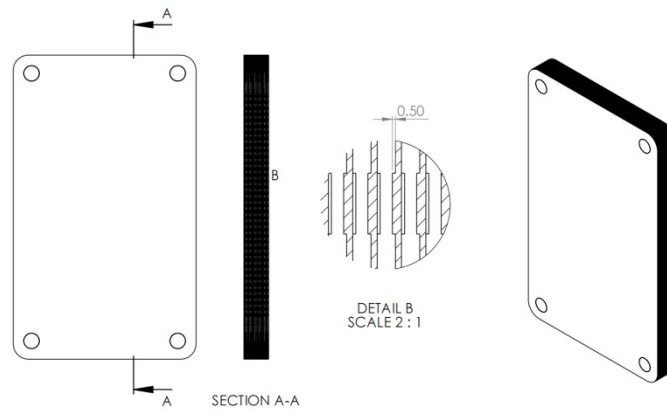
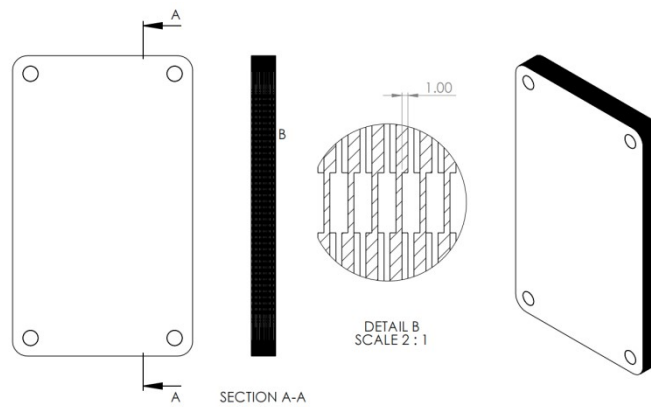


Figure 3.26 Cross sectional view of the cylindrical extrusion plate assembly



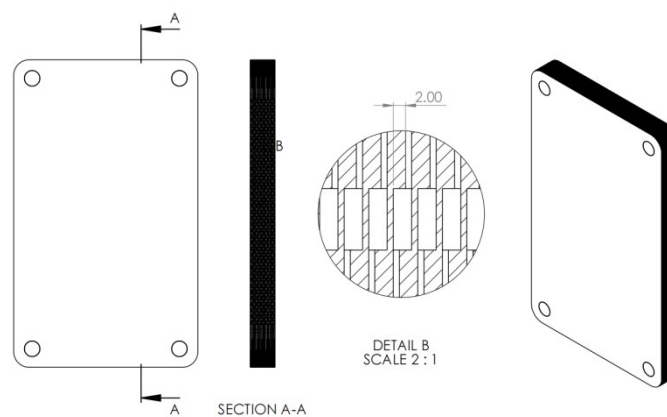
All dimensions are in 'mm'

Figure 3.27 Assembly of Plate with 0.5 mm cylindrical extrusion



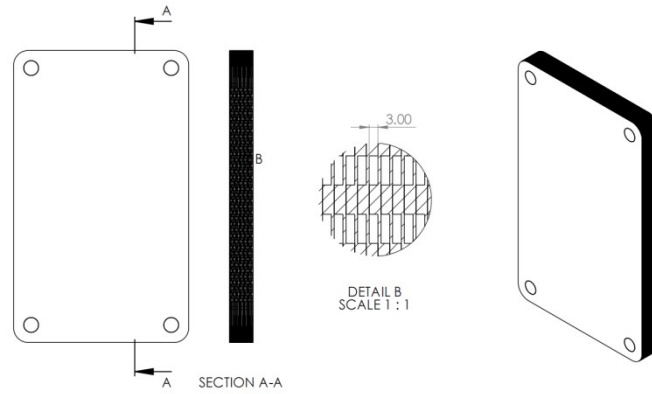
All dimensions are in 'mm'

Figure 3.28 Assembly of Plate with 1 mm cylindrical extrusion



All dimensions are in 'mm'

Figure 3.29 Assembly of Plate with 2 mm cylindrical extrusion



All dimensions are in 'mm'

Figure 3.30 Assembly of Plate with 3 mm cylindrical extrusion

3.3 Analysis of the Heat Exchanger

Cold fluid = MWCNT nanofluid

Hot fluid = Water

Design area of the plate, $A = L_p \times L_w = 439900 \text{ m}^2$

Mass flow rate of cold fluid, $m_{cf} = \rho_{nf} v_{cf} = 10.0922 \text{ kg/s}$

Mass flow rate of hot fluid, $m_{hf} = \rho_{nf} v_{hf} = 9.718 \text{ kg/s}$

Volume flow rate of cold fluid (v_{cf}) = Volume flow rate of hot fluid (v_{hf}) = $0.01 \text{ m}^3/\text{s}$

Temperature at inlet of cold fluid (T_{c1}) = 20°C

Temperature at inlet of hot fluid (T_{h1}) = 80°C

Heat transfer rate, $Q = m_{cf} C_{pnf} (T_{c2} - T_{c1}) = m_{hf} C_{pb} (T_{h1} - T_{h2})$

Effectiveness, $\epsilon = \{m_{cf} C_{pnf} (T_{c2} - T_{c1})\} / \{C_{\min}(T_{h1} - T_{c1})\}$

Pressure at exit (P_2) = 101.325 kPa

Pressure drop, $\Delta P = P_2 - P_1$

3.4 Summary of Design Parameters and Dimensions

Table 3.5 Design Parameters

Parameters	Values
Material of plate	Copper
Material of gasket	Teflon
Density of copper	8960 kg/m^3

Density of teflon	2160 kg/m ³
Diameter of the port	50 mm
Vertical distance between centre of ports	880 mm
Horizontal distance between ports	480 mm
Plate width inside gasket	530 mm
Plate length	830 mm
Chevron angle	30°
Mean channel spacing	3 mm
Corrugation pitch	20 mm
Plate thickness	1 mm
Number of plates	9
Number of gaskets	20
Total number of chambers	20
Number of hot fluid chambers	10
Number of cold fluid chambers	10
Design area of Plate	439900 mm ²
Volume flow rate of cold fluid	0.01 m ³ /s
Volume flow rate of hot fluid	0.01 m ³ /s
Mass flow rate of cold fluid	10.0922 kg/s
Mass flow rate of hot fluid	9.718 kg/s
Temperature at inlet of cold fluid	20°C
Temperature at inlet of hot fluid	80°C
Pressure at exit	101.325 kPa

CHAPTER 4: FLOW SIMULATION

4.1 System and General information

Table 4.1 System Info

Product	Flow Simulation 2023 SP5.0. Build: 6175
Computer name	USER_31
User name	Dell
Processors	Intel(R) Core(TM) i7-8700 CPU @ 3.20GHz
Memory	8015 MB / 3649 MB
Operating system	Windows 10 (Version 10.0.19041)
CAD version	SOLIDWORKS 2023 SP5.0
CPU speed	3192 MHz

Table 4.2 General Info

Model	-
Project name	Project(1)
Project comment	-
Project path	C:\Users\Dell\Desktop\ProjectME10\Model\
Units system	SI (m-k-g-s)
Analysis type	Internal
Exclude cavities without flow conditions	On
Coordinate system	Global Coordinate System
Reference axis	X

4.2 Physical Features

Fluid Flow: On

Conduction: On

Structural: Off

Electromagnetics: Off

Radiation: Off

Radiation in gases: Off

Time dependent: Off

Gravitational effects: Off

Rotation: Off

Flow type: Laminar and turbulent

High Mach number flow: Off

Free surface: Off

Default roughness: 1.0 micrometer

Default outer wall condition: Adiabatic wall

4.3 Initial Conditions

Thermodynamic parameters:

Ambient pressure = 101325.00 Pa

Ambient Temperature = 20.00 °C

Materials:

Solid: Copper, Teflon

Liquid: Water, MWCNT

Concentrations= 1, 1

Properties:

MWCNT

Density: 1009.22 kg/m³

Dynamic viscosity: 0.0027 Pa.s

Specific heat (C_p): 4109.3 J/(kg.K)

Thermal conductivity: 0.6690 W/(m.K)

Cavitation effect: No

Radiation properties: No

Water

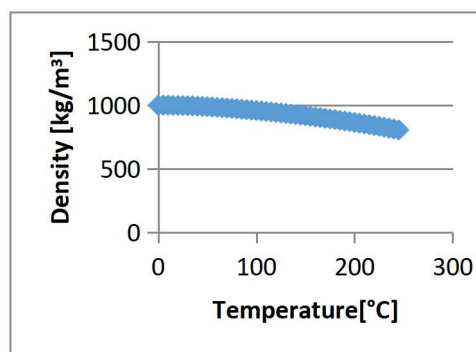


Figure 4.1 Density plot for water

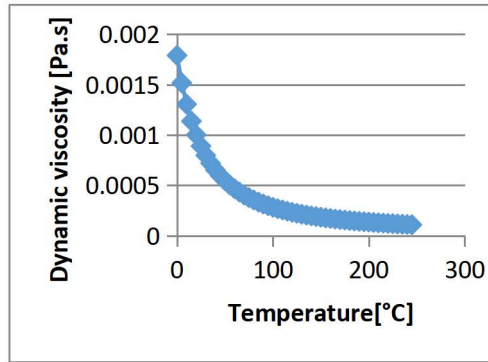


Figure 4.2 Dynamic viscosity plot for water

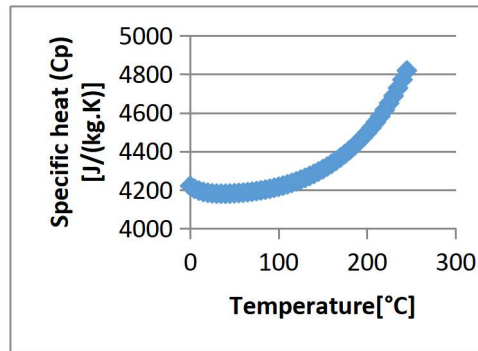


Figure 4.3 Specific heat plot for water

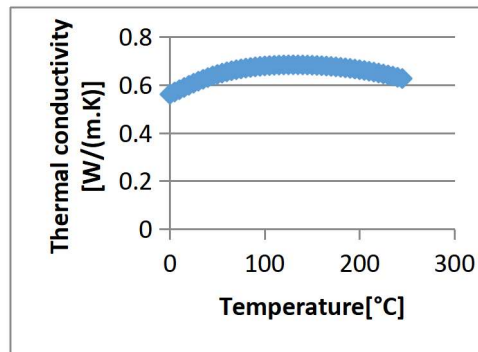


Figure 4.4 Thermal conductivity plot for water

4.4 Fluid Subdomains

Cold Fluid:

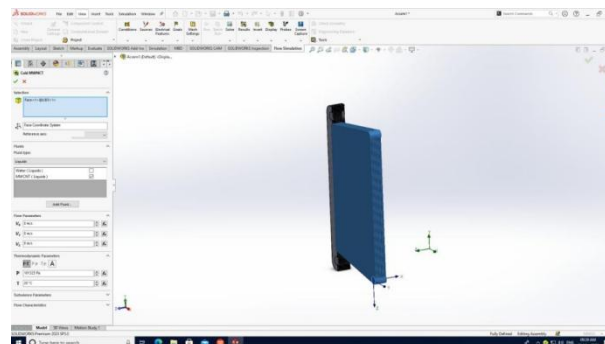


Figure 4.5 Cold MWCNT subdomain within the assembly

Fluid: Multi Walled Carbon Nanotube (MWCNT)

Static pressure = 101325 Pa

Fluid Temperature = 20.00 °C

Flow type: Laminar and turbulent

Turbulence intensity = 2%

Turbulence length = 8.1×10^{-4} m

Hot Fluid:

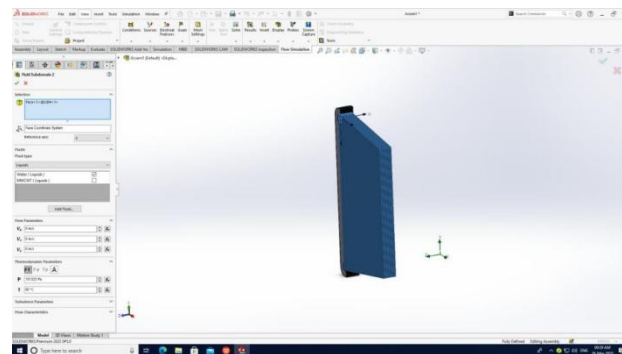


Figure 4.6 Hot water subdomain within the assembly

Fluid: Water

Static pressure = 101325 Pa

Fluid Temperature = 80.00 °C

Flow type: Laminar and turbulent

Turbulence intensity = 2%

Turbulence length = 8.1×10^{-4} m

4.5 Boundary Conditions

Inlet and outlet of the heat exchanger:

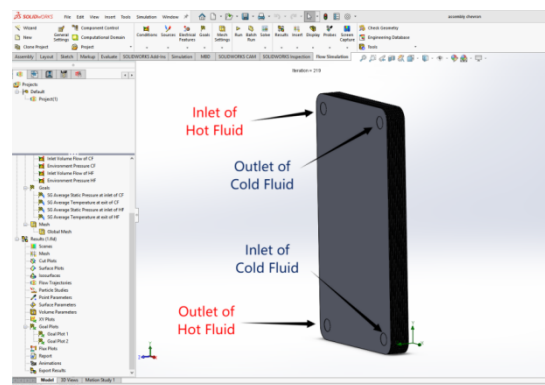


Figure 4.7 Inlet and outlet ports

Boundary conditions at:

Inlet of cold fluid: Volume flow rate = $0.01 \text{ m}^3/\text{s}$

Outlet of cold fluid: Environment pressure = 101325 Pa

Inlet of hot fluid: Volume flow rate = $0.01 \text{ m}^3/\text{s}$

Outlet of hot fluid: Environment pressure = 101325 Pa

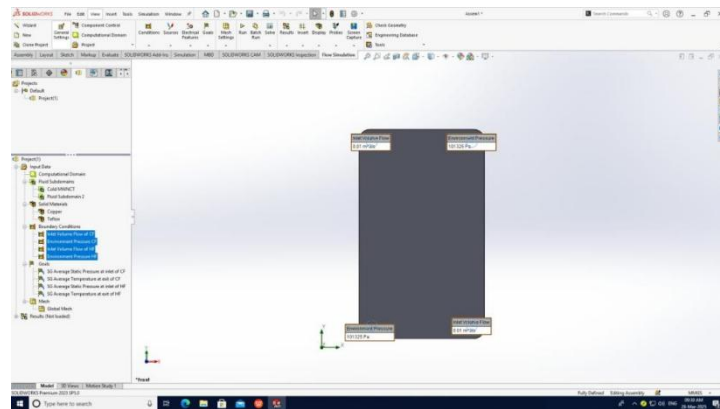


Figure 4.8 Boundary conditions at inlet and outlet ports

4.6 Goal Plots

Goals:

Goals type: Surface Goal

Inlet of cold fluid: Average static pressure

Outlet of cold fluid: Average fluid temperature

Inlet of hot fluid: Average static pressure

Outlet of hot fluid: Average fluid temperature

Table 4.3 Goal plot of flat plate

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
SG Average Static Pressure at inlet of CF	Pa	156832.14	100	857.657606	2.82771909	On

SG Average Static Pressure at exit of CF	Pa	101325.0 0	100	0.0010132 5	0	On
SG Average Temperature at exit of CF	°C	28.48	100	0.2181229 06	0.1537402 11	On
SG Average Static Pressure at inlet of HF	Pa	151803.4 3	100	595.29027 8	19.376711 7	On

Table 4.4 Goal plot of 0.5 mm chevron plate

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
SG Average Static Pressure at CF inlet	Pa	169202.4 9	100	747.92684 6	15.492139 6	On
SG Average Temperature at CF exit	°C	36.37	100	0.4262770 23	0.4181670 43	On
SG Average Static Pressure at HF inlet	Pa	155541.5 8	100	450.41546 1	3.6506072 8	On

SG Average Temperature at HF exit	°C	63.98	100	0.4212668 81	0.3215823 85	On
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Table 4.5 Goal plot of 1 mm chevron plate

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
SG Average Static Pressure at inlet of CF	Pa	259384.9 2	100	11267.871 3	374.87887 2	On
SG Average Temperature at exit of CF	°C	39.94	100	0.4912079 49	0.2973093 99	On
SG Average Static Pressure at inlet of HF	Pa	207678.4 0	100	7154.8502 7	27.485847 9	On
SG Average Temperature at exit of HF	°C	59.98	100	0.4775122 04	0.4695965 58	On

Table 4.6 Goal plot of 0.5 mm cylindrical extrusion plate

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
SG Average Static Pressure at CF inlet	Pa	160562. 32	100	809.4201 6	9.8409053 1	On
SG Average Temperatur e at CF exit	°C	32.38	100	0.316498 561	0.2727480 69	On
SG Average Static Pressure at HF inlet	Pa	148256. 07	100	494.0849 63	18.540583 4	On
SG Average Temperatur e at HF exit	°C	67.74	100	0.304984 143	0.3012978 92	On

Table 4.7 Goal plot of 1 mm cylindrical extrusion plate

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
SG Average Static Pressure at inlet of CF	Pa	163012 .54	100	724.69353 4	2.1955898 2	On
SG Average Temperatur e at exit of CF	°C	33.46	100	0.3426479 49	0.3375335 75	On

SG Average Static Pressure at inlet of HF	Pa	150739.52	100	485.232087	21.9550945	On
SG Average Temperature at exit of HF	°C	66.62	100	0.331436768	0.272579679	On

Table 4.8 Goal plot of 2 mm cylindrical extrusion plate

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
SG Average Static Pressure at CF inlet	Pa	170104.52	100	1355.99591	2.87816024	On
SG Average Temperature at CF outlet	°C	34.40	100	0.358916636	0.356793397	On
SG Average Static Pressure at HF inlet	Pa	155606.20	100	1055.03597	16.3029614	On
SG Average Temperature at HF outlet	°C	65.81	100	0.348694407	0.243604563	On

Table 4.9 Goal plot of 2.25 mm cylindrical extrusion plate

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
SG Average Static Pressure CF	Pa	171729. 05	100	1740.2343 2	6.380911 63	On
SG Average Temperatur e CF	°C	34.17	100	0.3495186 69	0.345290 659	On
SG Average Static Pressure HF	Pa	155875. 64	100	1377.7532 5	17.72196 03	On
SG Average Temperatur e HF	°C	66.00	100	0.3424330 58	0.223212 431	On

Table 4.10 Goal plot of 2.5 mm cylindrical extrusion plate

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
SG Average Static Pressure CF	Pa	173597.1 4	100	2082.6999 3	27.432592 3	On
SG Average Temperatur	°C	34.62	100	0.3600879 46	0.3542774 4	On

e CF						
SG Average Static Pressure HF	Pa	157346.4 6	100	1740.2747 7	19.647532 9	On
SG Average Temperatur e HF	°C	65.58	100	0.3521815 39	0.2265972 25	On

Table 4.11 Goal plot of 2.75 mm cylindrical extrusion plate

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
SG Average Static Pressure CF	Pa	175157.8 6	100	2296.0391 9	1.4311012	On
SG Average Temperatur e CF	°C	34.99	100	0.3702179 94	0.3660600 85	On
SG Average Static Pressure HF	Pa	158764.7 6	100	1914.4009 6	15.872336 4	On
SG Average Temperatur e HF	°C	65.21	100	0.3609865 26	0.2383366 35	On

Table 4.12 Goal plot of 3 mm cylindrical extrusion plate

Name	Unit	Value	Progress	Criteria	Delta	Use in convergence
SG Average Static Pressure at CF inlet	Pa	177753.28	100	1838.7438	9.63201676	On
SG Average Temperature at CF exit	°C	34.20	100	0.342810622	0.337884259	On
SG Average Static Pressure at HF inlet	Pa	161016.77	100	1498.30527	15.741239	On
SG Average Temperature at HF exit	°C	65.73	100	0.343680131	0.186493823	On

CHAPTER 5: RESULTS AND DISCUSSIONS

5.1 Initial Results

5.1.1 Initial observed data from simulation

Table 5.1 Unknown pressure and temperature values obtained from simulation

Plate type	Cold fluid		Hot Fluid	
	Exit temperature (T_{c2}) °C	Inlet pressure (P_{c1}) kPa	Exit temperature (T_{h2}) °C	Inlet pressure (P_{h2}) kPa
Flat Plate	28.48	156.838	72.29	151.780
0.5 mm chevron plate	33.46	203.012	66.62	182.740
1 mm chevron plate	39.903	259.231	60.11	207.701
0.5 mm cylindrical extrusion plate	32.28	160.501	67.85	148.250
1 mm cylindrical extrusion plate	33.33	163.015	66.73	150.723
2 mm cylindrical extrusion plate	34.40	170.106	65.91	155.618
3 mm cylindrical extrusion plate	34.06	177.768	65.79	161.017

5.1.2 Calculation of heat transfer rate and pressure drop

Table 5.2 Heat transfer, effectiveness and pressure drop for all the plates

Plate type	Heat transfer rate (kW)	Effectiveness	Pressure drop (kPa)
Flat Plate	351.684	0.142	55.513
0.5 mm chevron plate	558.217	0.224	101.687
1 mm chevron plate	825.465	0.332	157.096
0.5 mm cylindrical extrusion plate	509.475	0.206	59.208
1 mm cylindrical extrusion plate	552.766	0.218	61.69
2 mm cylindrical extrusion plate	591.026	0.257	68.766
3 mm cylindrical extrusion plate	586.833	0.240	76.443

5.1.3 Comparison of chevron plate with flat plate

Table 5.3 Comparison of chevron plate with flat plate

Plate pitch (mm)	Increase in heat transfer rate (kW)	% increase in heat transfer rate	Increase in pressure drop (kPa)	% increase in pressure drop
0.5	206.53	58.73 %	46.174	83.18 %
1	473.78	134.72 %	101.58	182.99 %

5.1.4 Comparison of cylindrical extrusion plate with flat plate

Table 5.4 Comparison of cylindrical extrusion plate with flat plate

Height of extrusion (mm)	Increase in heat transfer rate (kW)	% increase in heat transfer rate	Increase in pressure drop (kPa)	% increase in pressure drop
0.5	157.79	44.87 %	3.695	6.66 %
1	201.08	57.18 %	6.18	11.13 %
2	239.34	68.05 %	13.25	23.87 %
3	235.15	66.86 %	20.93	37.7 %

5.1.5 Comparison of cylindrical extrusion plate with chevron plate

Table 5.5 Comparison of cylindrical extrusion plate with chevron plate

Height of extrusion (mm)	Chevron plate (0.5 mm plate pitch)				Chevron plate (1 mm plate pitch)			
	Decrease in heat transfer rate (kW)	% decrease in heat transfer rate	Decrease in pressure drop (kPa)	% decrease in pressure drop	Decrease in heat transfer rate (kW)	% decrease in heat transfer rate	Decrease in pressure drop (kPa)	% decrease in pressure drop
0.5	48.742	8.73%	42.48	41.77 %	315.99	38.28 %	97.89	62.31 %
1	5.451	0.98%	40	39.33 %	272.7	33.04 %	95.41	60.73 %
2	-32.81	-5.88%	32.921	32.37 %	234.44	28.4%	88.33	56.23 %
3	-28.62	-5.13%	25.24	24.83 %	238.63	28.91 %	80.65	51.34 %

5.1.6 Results

1. The flat plate serves as the baseline, showing the lowest heat transfer rate of 351.684 kW and also the lowest pressure drop of 55.513 kPa. It has the minimum pressure drop among all configuration but also shows very limited heat transfer capability.

2. The chevron plate with 1 mm plate pitch has the maximum heat transfer rate (825.465 kW) with an increase of 134.72 % to that of the flat plate. But the pressure drop associated with it (157.096 kPa) is also maximum with an increase of 182.9%.

3. For the chevron plate with 0.5 mm plate pitch the heat transfer rate is slightly less (with an increase of 58.73% to that of flat plate). However the pressure drop is still significant with an increase of 83.18%.

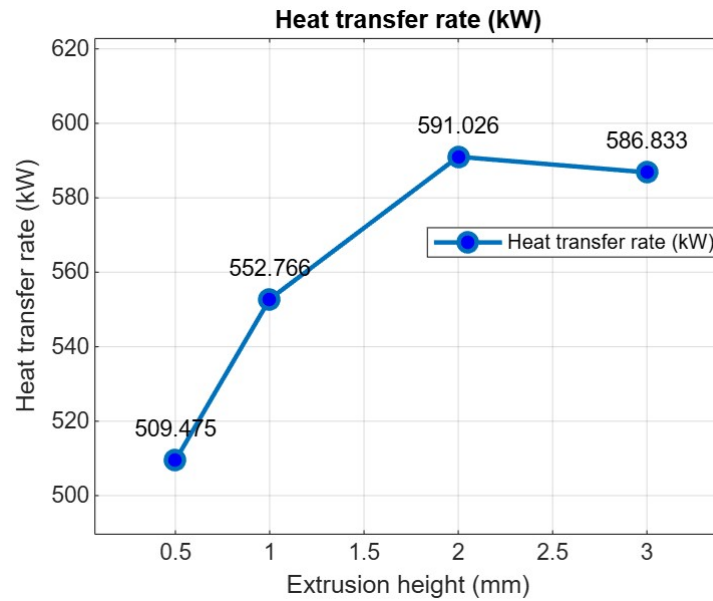


Figure 5.1 Plot of heat transfer rate vs height of extrusion

4. In case of the cylindrical extrusion plates, the heat transfer increases with the increase in extrusion height. But for height 3 mm there is a slight reduction in heat transfer rate (almost 4.2 kW) compared to 2 mm extrusion. Since the gap between the plates is limited to 3 mm, therefore it shows that the optimal heat transfer lies between 2 mm to 3 mm extrusion height.

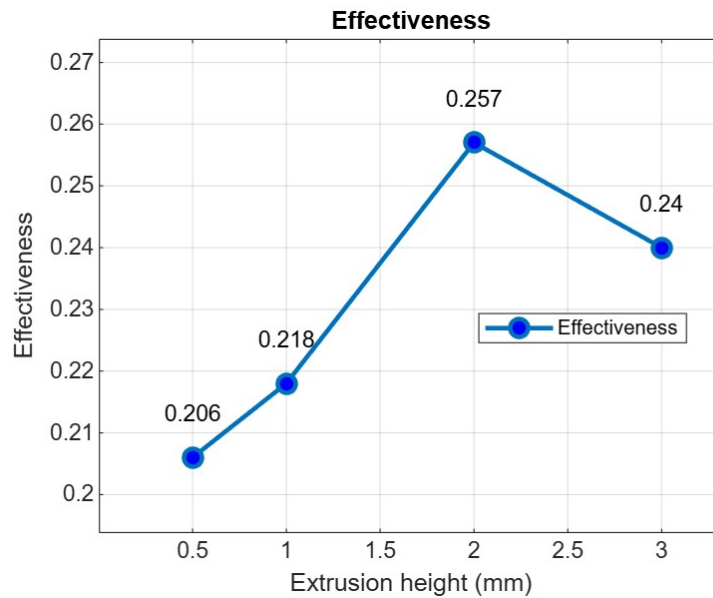


Figure 5.2 Plot of effectiveness vs height of extrusion

5. The effectiveness also follows the same trend as that of the heat transfer rate. It also drops for extrusion height of 3 mm.

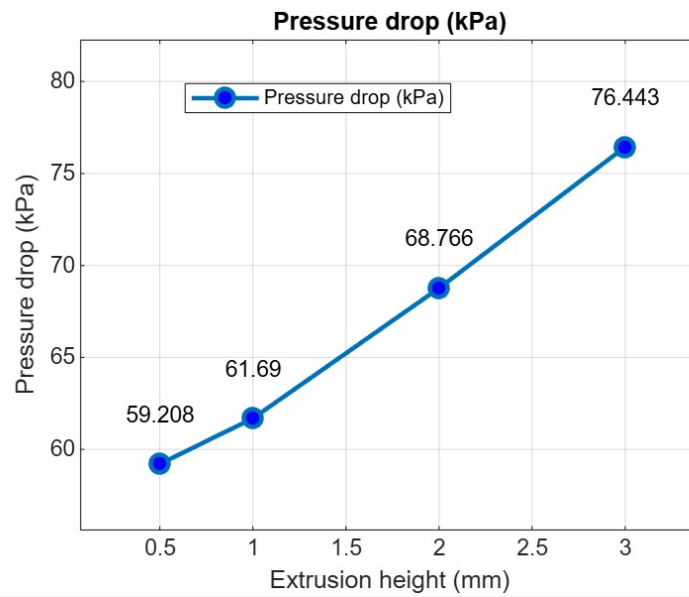


Figure 5.3 Plot of pressure drop vs height of extrusion

6. But the trend of increase in pressure drop is maintained throughout all heights of the extrusions. It increases with the increase in extrusion height and is maximum for 3 mm extrusion (76.443 kPa).

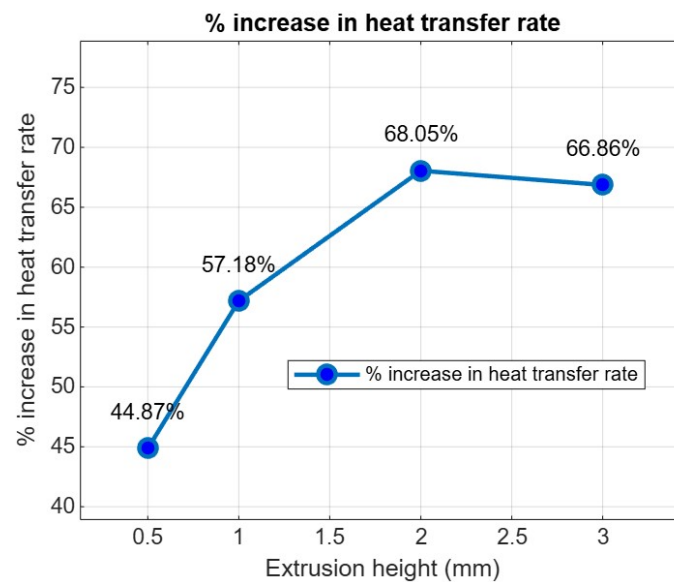


Figure 5.4 Plot of % increase in heat transfer rate vs height of extrusion

7. In comparison to flat plate the increase in heat transfer rate increase with extrusion height reaching peak at 2 mm (68.05%) and then decreasing slightly for 3 mm height.

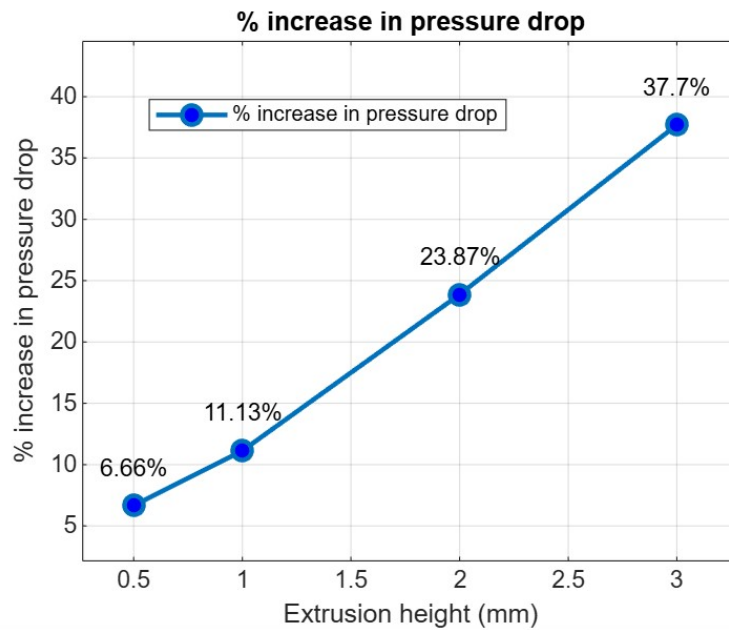


Figure 5.5 Plot of % increase in pressure drop vs height of extrusion

8. The increase in pressure drop for cylindrical extrusion plates is very less with maximum of 37.7% only. The heat transfer enhancement is significant over this slight increase pressure drop.

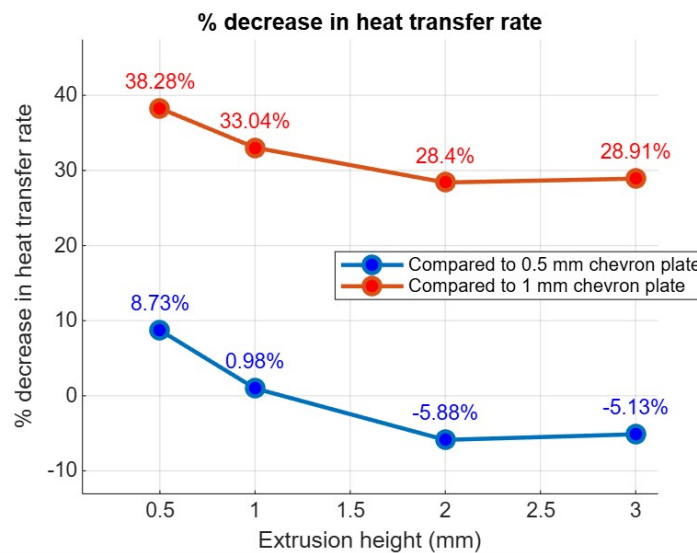


Figure 5.6 Plot of % decrease in heat transfer rate vs height of extrusion

9. In comparison to 0.5 mm chevron plate, the reduction in heat transfer for cylindrical extrusion plates are very negligible. At 2 mm and 3 mm extrusion, the heat

transfer is more compared to 0.5 mm chevron plate. 1 mm chevron plate is proved to have higher heat transfer rate throughout.

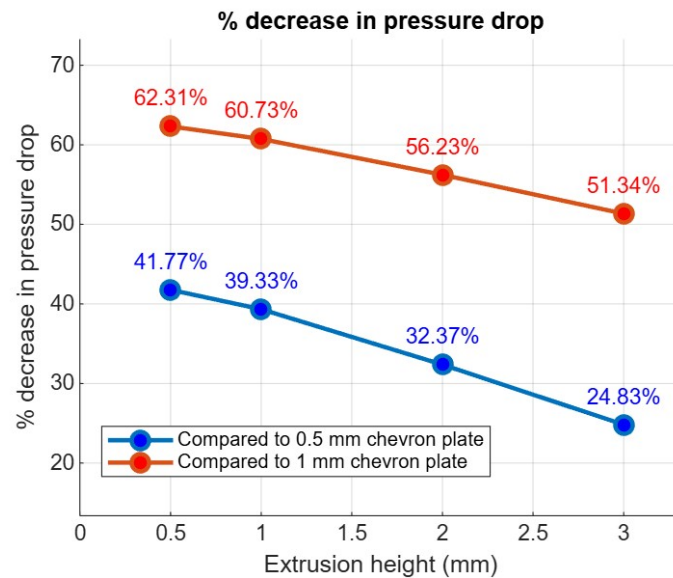


Figure 5.7 Plot of % decrease in pressure drop vs height of extrusion

10. The pressure drop comes out to be very less in case of extrusion plates in comparison to chevron plates. The decrease in pressure drop is significantly high for all values of extrusion height.

5.1.7 Summary of initial results

Plates with cylindrical extrusions demonstrate a balance between heat transfer enhancement and pressure drop reduction. As the extrusion height increases, the heat transfer rate improves, but the pressure drop also rises. The heat transfer however decreases for 3 mm extrusion. For example: The 0.5 mm extrusion provides a modest heat transfer improvement (509.475 kW, 44.87% higher than the flat plate) with a negligible pressure drop increase of 6.66%. The 2 mm extrusion achieves the best trade-off among cylindrical plates, with a 68.05% increase in heat transfer (591.026 kW) and a moderate pressure drop increase of 23.87% (68.766 kPa). The 3 mm extrusion slightly reduces heat transfer (586.833 kW) while increasing the pressure drop to 76.443 kPa, indicating diminishing returns for taller extrusions. While cylindrical extrusion plates do not match chevron plates in heat transfer performance, they significantly reduce pressure drops. For example the 0.5 mm extrusion plate

reduces pressure drop by 41.77% compared to the 0.5 mm chevron plate, with only an 8.73% decrease in heat transfer. This trend holds across all extrusion heights. In comparison to the 0.5 mm chevron plate the heat transfer rate is more in cylindrical extrusion with height more than 1 mm. In comparison to the 1 mm chevron plate also the trend of decline in the decrease of heat transfer rate as well as pressure drop is maintained over the increasing heights of extrusions.

5.2 Further simulation

Five results for each type of plate (flat plate, 0.5 mm, 1 mm, 2 mm, 2.25 mm, 2.50 mm, 2.75 mm and 3 mm cylindrical extrusion plates) is further taken to increase the sample size for analysis of variance testing.

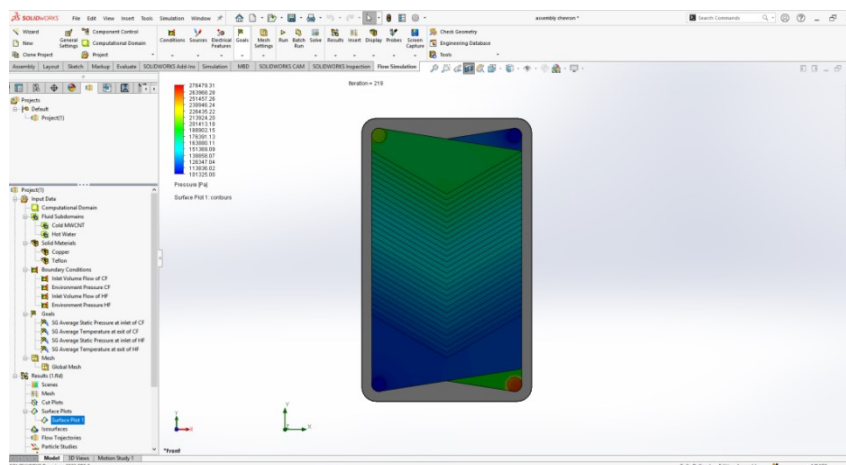


Figure 5.8 Pressure distribution profile

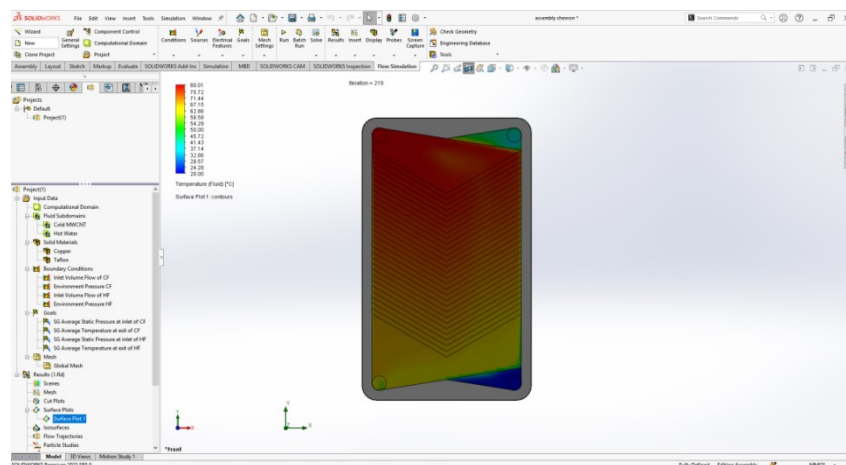


Figure 5.9 Fluid temperature distribution profile

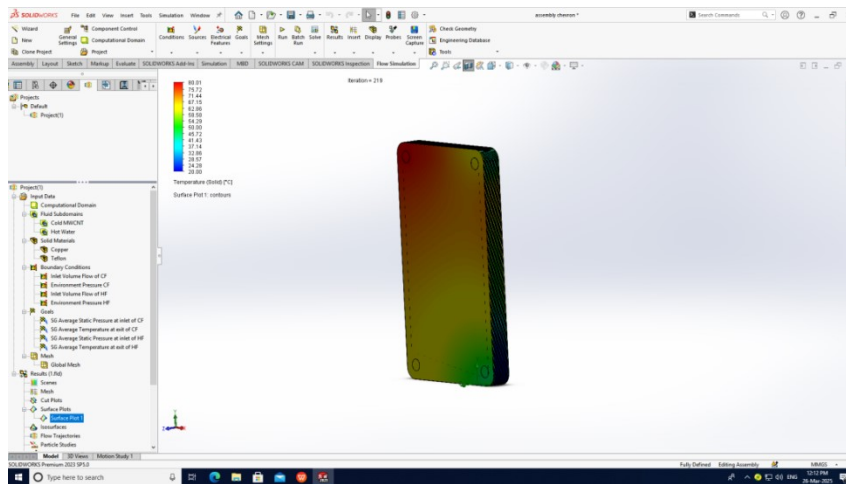


Figure 5.10 Solid temperature distribution profile

5.2.1 Repeated results for all plate types

The inlet pressures and the exit temperatures of the five repeated simulations for both the hot fluid and the cold fluid are obtained for all plate types to calculate the corresponding heat transfer and pressure drop values on which anova testing can be done.

5.2.1.1 Flat plate

Table 5.6 Results for flat plate

Sl No	Inlet pressure of cold fluid (kPa)	Exit temperature of cold fluid (°C)	Inlet pressure of hot fluid (kPa)	Exit temperature of hot fluid (°C)
1	156.832	28.480	151.803	72.210
2	155.769	28.4864	144.978	72.0455
3	155.769	28.4865	144.920	72.0458
4	155.769	28.4864	144.917	72.0455
5	155.769	28.4865	144.920	72.0458
Average	155.9816	28.48516	146.3076	72.07852

5.2.1.2 Cylindrical extrusion plate of 0.5 mm

Table 5.7 Results for cylindrical extrusion plate of 0.5 mm

Sl No	Inlet pressure of cold fluid (kPa)	Exit temperature of cold fluid (°C)	Inlet pressure of hot fluid (kPa)	Exit temperature of hot fluid (°C)
1	160.501	32.2800	148.250	67.850
2	158.858	31.8888	147.156	68.2977
3	158.858	31.8889	147.156	68.2978
4	158.858	31.8887	147.156	68.2978
5	158.858	31.8887	147.156	68.2978
Average	159.1866	31.96702	147.3748	68.20822

5.2.1.3 Cylindrical extrusion plate of 1 mm

Table 5.8 Results for cylindrical extrusion plate of 1 mm

Sl No	Inlet pressure of cold fluid (kPa)	Exit temperature of cold fluid (°C)	Inlet pressure of hot fluid (kPa)	Exit temperature of hot fluid (°C)
1	163.015	33.330	150.723	66.730
2	162.057	32.9104	149.324	67.1591
3	162.057	32.9105	149.324	67.159
4	162.057	32.9105	149.324	67.159
5	162.057	32.9104	149.324	67.1591
Average	162.2486	32.99436	149.6038	67.07324

5.2.1.4 Cylindrical extrusion plate of 2 mm

Table 5.9 Results for cylindrical extrusion plate of 2 mm

Sl No	Inlet pressure of cold fluid (kPa)	Exit temperature of cold fluid (°C)	Inlet pressure of hot fluid (kPa)	Exit temperature of hot fluid (°C)
1	170.106	34.400	155.618	65.91
2	169.604	33.7548	154.399	66.3695
3	169.602	33.7546	154.399	66.3694
4	169.603	33.7547	154.399	66.3695
5	169.604	33.7548	154.399	66.3695
Average	169.7038	33.88378	154.6428	66.27758

5.2.1.5 Cylindrical extrusion plate of 2.25 mm

Table 5.10 Results for cylindrical extrusion plate of 2.25 mm

Sl No	Inlet pressure of cold fluid (kPa)	Exit temperature of cold fluid (°C)	Inlet pressure of hot fluid (kPa)	Exit temperature of hot fluid (°C)
1	171.675	34.0537	155.885	66.0745
2	171.712	34.0253	155.884	66.0937
3	171.712	34.0254	155.887	66.0935
4	171.703	34.0252	155.886	66.0936
5	171.709	34.0251	155.885	66.0938
Average	171.7022	34.03094	155.8854	66.08982

5.2.1.6 Cylindrical extrusion plate of 2.50 mm

Table 5.11 Results for cylindrical extrusion plate of 2.50 mm

Sl No	Inlet pressure of cold fluid (kPa)	Exit temperature of cold fluid (°C)	Inlet pressure of hot fluid (kPa)	Exit temperature of hot fluid (°C)
1	173.609	34.475	157.355	65.6671
2	173.610	34.475	157.354	65.6669
3	173.608	34.4749	157.355	65.6671
4	173.608	34.4749	157.354	65.6672
5	173.608	34.4749	157.354	65.6675
Average	173.6086	34.47494	157.3544	65.66716

5.2.1.7 Cylindrical extrusion plate of 2.75 mm

Table 5.12 Results for cylindrical extrusion plate of 2.75 mm

Sl No	Inlet pressure of cold fluid (kPa)	Exit temperature of cold fluid (°C)	Inlet pressure of hot fluid (kPa)	Exit temperature of hot fluid (°C)
1	175.157	34.8425	158.767	65.3067
2	175.158	34.8432	158.767	65.3057
3	175.158	34.8422	158.769	65.3068
4	175.157	34.8427	158.767	65.3069
5	175.157	34.8422	158.766	65.3064
Average	175.1574	34.84256	158.7672	65.3065

5.2.1.8 Cylindrical extrusion plate of 3 mm

Table 5.13 Results for cylindrical extrusion plate of 3 mm

Sl No	Inlet pressure of cold fluid (kPa)	Exit temperature of cold fluid (°C)	Inlet pressure of hot fluid (kPa)	Exit temperature of hot fluid (°C)
1	177.768	34.0600	161.017	65.7900
2	176.810	33.9346	159.700	65.9201
3	176.809	33.9334	159.701	65.9198
4	176.810	33.9333	159.700	65.9198
5	176.810	33.9343	159.967	65.9190
Average	177.0014	33.95912	160.017	65.89374

5.2.2 Heat transfer rate and Pressure drop values

Table 5.14 Heat transfer rates and pressure drops for all plates

Plate Type	Extrusion height	Sl No.	Heat transfer rate (kW)	Pressure drop (kPa)
Flat Plate	N/A	1	351.6849	55.5070
		2	351.9504	53.8440
		3	351.9545	53.8440
		4	351.9504	53.8440
		5	351.9505	53.8440
	0.5 mm	1	509.2796	59.1760
		2	493.0557	57.5330
		3	493.0598	57.5330
		4	493.0515	57.5330
		5	493.0515	57.5330

Plate with cylindrical extrusions	1 mm	1	552.8255	61.6900
		2	535.4237	60.7320
		3	535.4279	60.7320
		4	535.4279	60.7320
		5	535.4237	60.7320
	2 mm	1	597.2008	68.7810
		2	570.4429	68.2790
		3	570.4346	68.2770
		4	570.4388	68.2780
		5	570.4429	68.2790
	2.25 mm	1	582.8390	70.3500
		2	581.6612	70.3870
		3	581.6653	70.3870
		4	581.6570	70.3780
		5	581.6529	70.3840
	2.50 mm	1	600.3113	72.2840
		2	600.3113	72.2850
		3	600.3071	72.2830
		4	600.3071	72.2830
		5	600.3071	72.2830
	2.75 mm	1	615.5523	73.8320
		2	615.5814	73.8330
		3	615.5399	73.8330
		4	615.5606	73.8320
		5	615.5399	73.8320
	3 mm	1	583.1003	76.4430
		2	577.8996	75.4850
		3	577.8499	75.4840
		4	577.8457	75.4850
		5	577.8872	75.4850

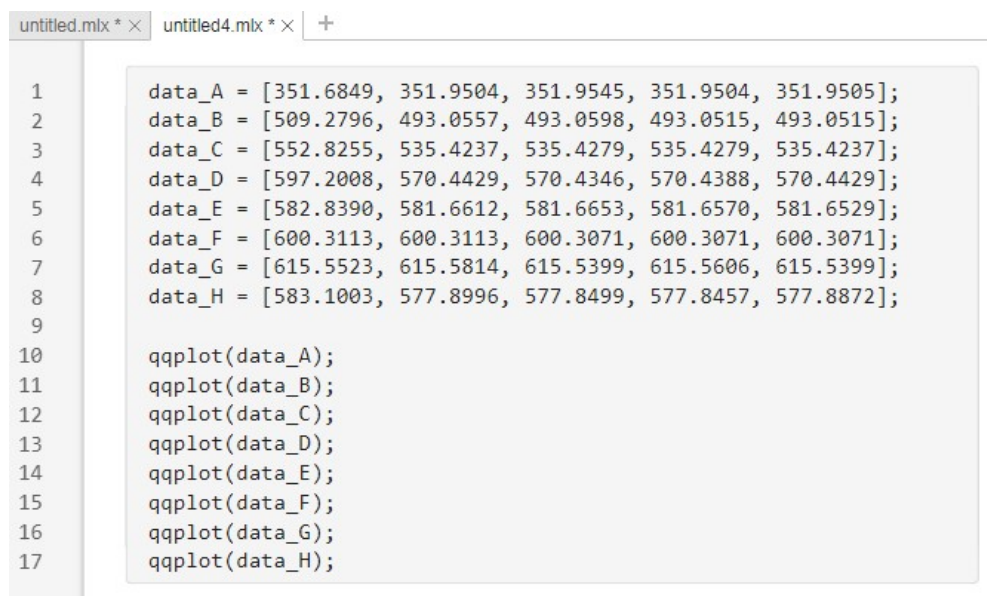
5.3 Normality check for the data

The heat transfer rate and pressure drop data are checked whether it can be assumed to be normally distributed or not by using Quantile-Quantile (Q-Q) plots for each groups. Further for analytical verification of the plot result Anderson-Darling test is done.

The Q-Q plot is a plot between a theoretical quantile (x-axis) and empirical quantiles (y-axis). The theoretical quantiles are obtained from the standard normal distribution and the empirical quantiles corresponds to the sample data. A straight line is plotted between these two and it represents the points which are perfectly normally distributed. If the sample data points lies on or near the straight line then the sample is assumed to be normally distributed and vice versa.

5.3.1 Q-Q plots for heat transfer rate

For plotting the Q-Q plots the built in 'qqplot(group)' of matlab is used. The following code is used to get the Q-Q plot of all the eight groups. Here, group A, B, C, D, E, F, G, H corresponds to the groups of flat plate and cylindrical extrusion plate of 0.5 mm, 1mm, 2 mm, 2.25 mm, 2.50 mm, 2.75 mm & 3 mm respectively.

The image shows a MATLAB code editor window with two tabs: 'untitled.mlx' and 'untitled4.mlx'. The code is written in the 'untitled4.mlx' tab. It consists of two parts: data assignment and plotting. Lines 1-8 assign data arrays 'data_A' through 'data_H' with specific numerical values. Lines 10-17 use the 'qqplot' function to create Q-Q plots for each of these data arrays. The code is as follows:

```
1 data_A = [351.6849, 351.9504, 351.9545, 351.9504, 351.9505];
2 data_B = [509.2796, 493.0557, 493.0598, 493.0515, 493.0515];
3 data_C = [552.8255, 535.4237, 535.4279, 535.4279, 535.4237];
4 data_D = [597.2008, 570.4429, 570.4346, 570.4388, 570.4429];
5 data_E = [582.8390, 581.6612, 581.6653, 581.6570, 581.6529];
6 data_F = [600.3113, 600.3113, 600.3071, 600.3071, 600.3071];
7 data_G = [615.5523, 615.5814, 615.5399, 615.5606, 615.5399];
8 data_H = [583.1003, 577.8996, 577.8499, 577.8457, 577.8872];
9
10 qqplot(data_A);
11 qqplot(data_B);
12 qqplot(data_C);
13 qqplot(data_D);
14 qqplot(data_E);
15 qqplot(data_F);
16 qqplot(data_G);
17 qqplot(data_H);
```

Figure 5.11 Code for plotting the Q-Q plot for heat transfer rate data

The output plots are:

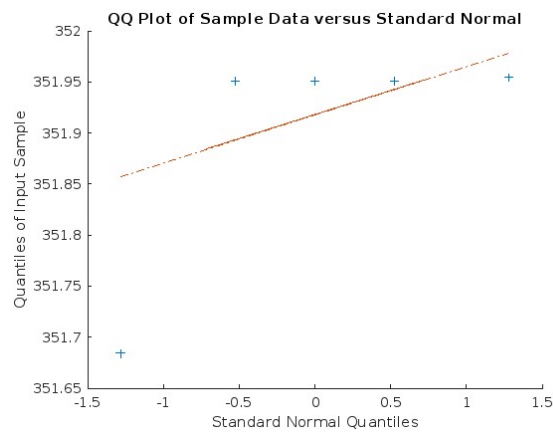


Figure 5.12 Q-Q plot for heat transfer data for group A

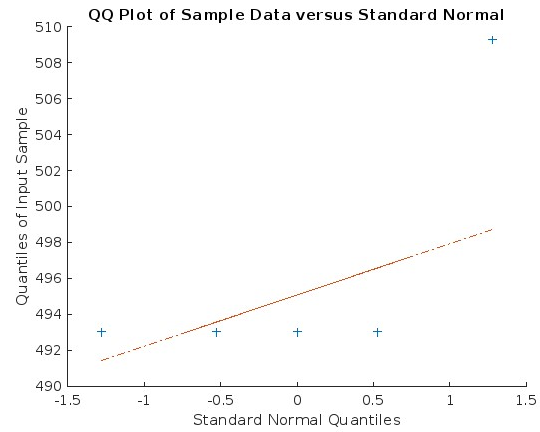


Figure 5.13 Q-Q plot for heat transfer data for group B

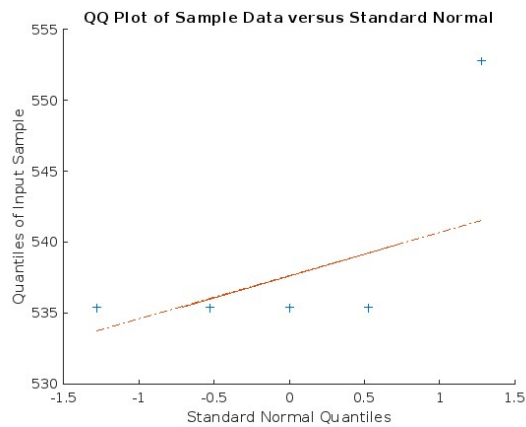


Figure 5.14 Q-Q plot for heat transfer data for group C

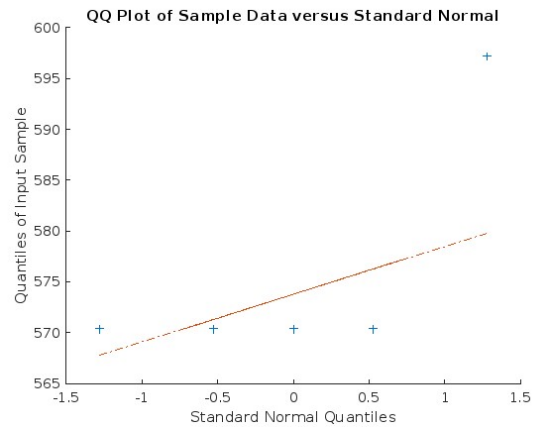


Figure 5.15 Q-Q plot for heat transfer data for group D

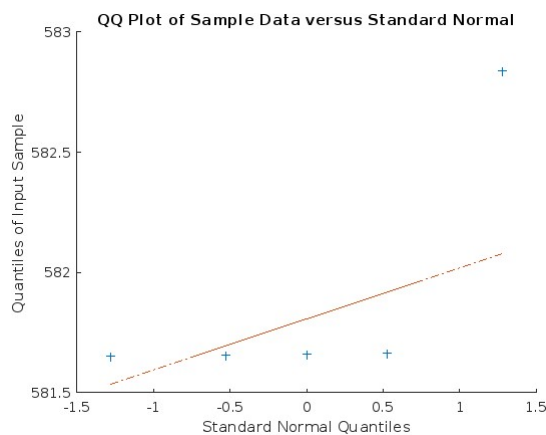


Figure 5.16 Q-Q plot for heat transfer data for group E

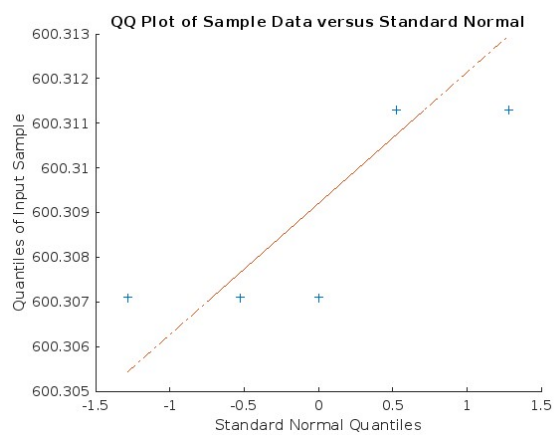


Figure 5.17 Q-Q plot for heat transfer data for group F

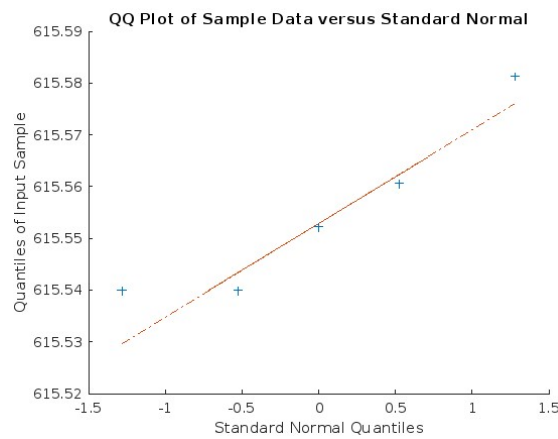


Figure 5.18 Q-Q plot for heat transfer data for group G

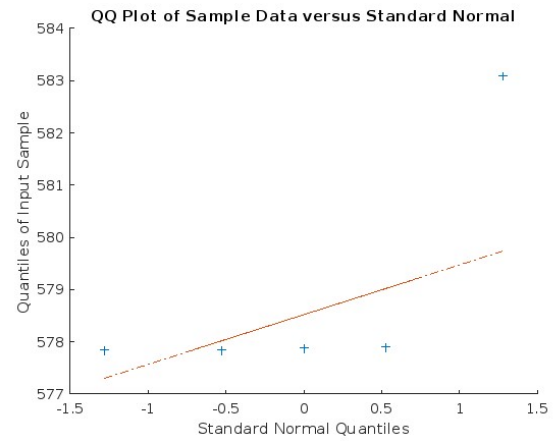


Figure 5.19 Q-Q plot for heat transfer data for group H

5.3.2 Q-Q plot for pressure drop

Similarly the Q-Q plots for pressure drop are obtained with the following code.

```

untitled.mlx * ×  untitled4.mlx * ×  untitled7.mlx * ×  +
1      A = [55.5070, 53.8440, 53.8440, 53.8440, 53.8440];
2      B = [59.1760, 57.5330, 57.5330, 57.5330, 57.5330];
3      C = [61.6900, 60.7320, 60.7320, 60.7320, 60.7320];
4      D = [68.7810, 68.2790, 68.2770, 68.2780, 68.2790];
5      E = [70.3500, 70.3870, 70.3870, 70.3780, 70.3840];
6      F = [72.2840, 72.2850, 72.2830, 72.2830, 72.2830];
7      G = [73.8320, 73.8330, 73.8330, 73.8320, 73.8320];
8      H = [76.4430, 75.4850, 75.4840, 75.4850, 75.4850];
9
10     qqplot(A);
11     qqplot(B);
12     qqplot(C);
13     qqplot(D);
14     qqplot(E);
15     qqplot(F);
16     qqplot(G);
17     qqplot(H);

```

Figure 5.20 Code for plotting the Q-Q plot for pressure drop data

The output plots are:

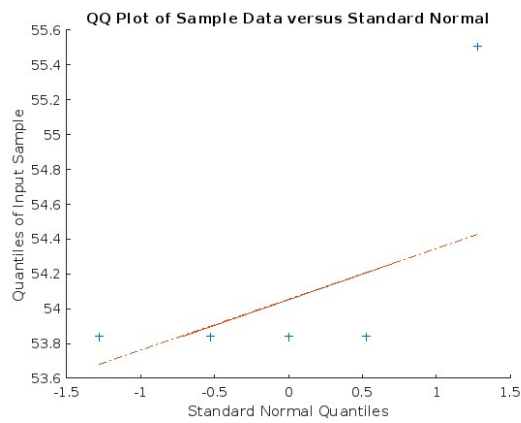


Figure 5.21 Q-Q plot for pressure drop data for group A

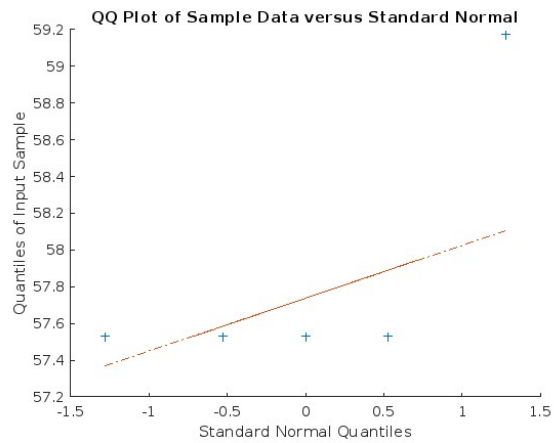


Figure 5.22 Q-Q plot for pressure drop data for group B

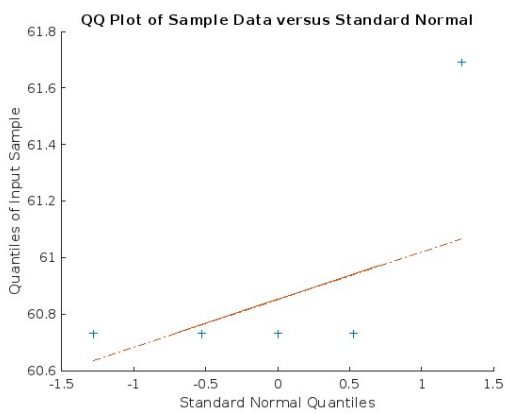


Figure 5.23 Q-Q plot for pressure drop data for group C

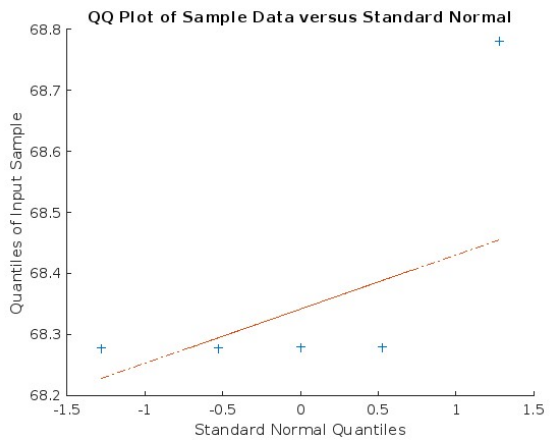


Figure 5.24 Q-Q plot for pressure drop data for group D

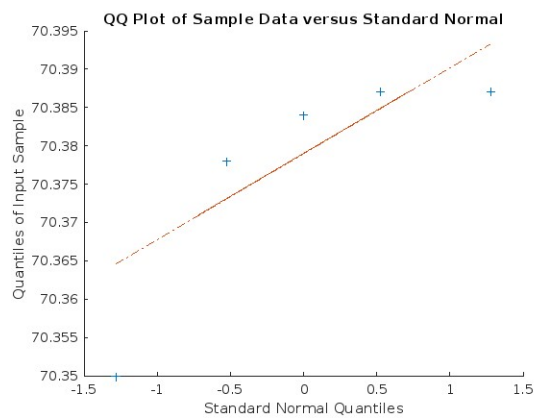


Figure 5.25 Q-Q plot for pressure drop data for group E

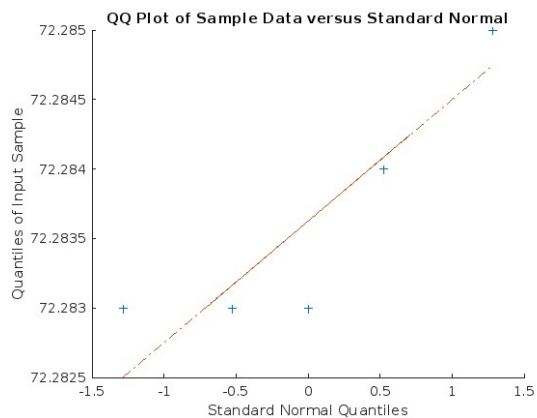


Figure 5.26 Q-Q plot for pressure drop data for group F

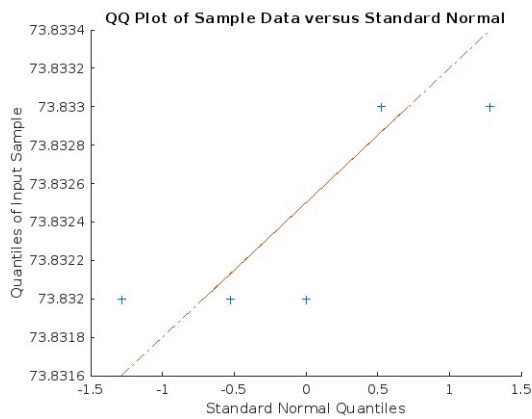


Figure 5.27 Q-Q plot for pressure drop data for group G

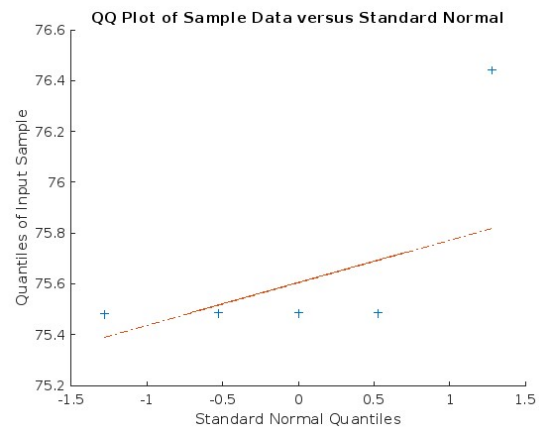


Figure 5.28 Q-Q plot for pressure drop data for group H

From the above Q-Q plots it is observed that the sample data points are spread significantly far away from the normal straight line. Therefore it indicates that the data in all the groups can not be assumed to be normally distributed.

5.3.3 Anderson-Darling test

Anderson-Darling test is done to quantify and further verify the results of the Q-Q plots. The code for the test for heat transfer rates and pressure drops respectively are as follows.

```
untitled.mlx × +
1 A = [351.6849, 351.9504, 351.9545, 351.9504, 351.9505];
2 B = [509.2796, 493.0557, 493.0598, 493.0515, 493.0515];
3 C = [552.8255, 535.4237, 535.4279, 535.4279, 535.4237];
4 D = [597.2008, 570.4429, 570.4346, 570.4388, 570.4429];
5 E = [582.8390, 581.6612, 581.6653, 581.6570, 581.6529];
6 F = [600.3113, 600.3113, 600.3071, 600.3071, 600.3071];
7 G = [615.5523, 615.5814, 615.5399, 615.5606, 615.5399];
8 H = [583.1003, 577.8996, 577.8499, 577.8457, 577.8872];
9
10 [h, p, ads] = adtest(A);
11
12 disp(['Anderson-Darling Test for Group A:']);
13 disp(['h = ', num2str(h), ', p = ', num2str(p), ', AD Statistic = ', num2str(ads)]);
14
```

Figure 5.29 Code for Anderson-Darling test for heat transfer rate

```

1 A = [55.5070, 53.8440, 53.8440, 53.8440, 53.8440];
2 B = [59.1760, 57.5330, 57.5330, 57.5330, 57.5330];
3 C = [61.6900, 60.7320, 60.7320, 60.7320, 60.7320];
4 D = [68.7810, 68.2790, 68.2770, 68.2780, 68.2790];
5 E = [70.3500, 70.3870, 70.3870, 70.3780, 70.3840];
6 F = [72.2840, 72.2850, 72.2830, 72.2830, 72.2830];
7 G = [73.8320, 73.8330, 73.8330, 73.8320, 73.8320];
8 H = [76.4430, 75.4850, 75.4840, 75.4850, 75.4850];
9
10 [h, p, ads] = adtest(A);
11
12 disp(['Anderson-Darling Test for Group A:']);
13 disp(['h = ', num2str(h), ', p = ', num2str(p), ', AD Statistic = ', num2str(ads)]);
14

```

Figure 5.30 Code for Anderson-Darling test for pressure drop

The output for the test is:

Warning: P is less than the smallest tabulated value, returning 0.0005.

Anderson-Darling Test for Group A:

h = 1, p = 0.0005, AD Statistic = 1.171

Warning: P is less than the smallest tabulated value, returning 0.0005.

Anderson-Darling Test for Group B:

h = 1, p = 0.0005, AD Statistic = 1.2034

Warning: P is less than the smallest tabulated value, returning 0.0005.

Anderson-Darling Test for Group C:

h = 1, p = 0.0005, AD Statistic = 1.204

Warning: P is less than the smallest tabulated value, returning 0.0005.

Anderson-Darling Test for Group D:

h = 1, p = 0.0005, AD Statistic = 1.2039

Anderson-Darling Test for Group E:

h = 1, p = 0.0005, AD Statistic = 1.1791

Anderson-Darling Test for Group F:

h = 1, p = 0.0091747, AD Statistic = 0.79955

Anderson-Darling Test for Group G:

h = 0, p = 0.43676, AD Statistic = 0.31633

Warning: P is less than the smallest tabulated value, returning 0.0005.

Anderson-Darling Test for Group H:

h = 1, p = 0.0005, AD Statistic = 1.177

Figure 5.31 Output for A-D test for heat transfer rate

```

Warning: P is less than the smallest tabulated value, returning 0.0005.

Anderson-Darling Test for Group A:
h = 1, p = 0.0005, AD Statistic = 1.2047
Warning: P is less than the smallest tabulated value, returning 0.0005.

Anderson-Darling Test for Group B:
h = 1, p = 0.0005, AD Statistic = 1.2047
Warning: P is less than the smallest tabulated value, returning 0.0005.

Anderson-Darling Test for Group C:
h = 1, p = 0.0005, AD Statistic = 1.2047
Warning: P is less than the smallest tabulated value, returning 0.0005.

Anderson-Darling Test for Group D:
h = 1, p = 0.0005, AD Statistic = 1.1945

Anderson-Darling Test for Group E:
h = 1, p = 0.024844, AD Statistic = 0.68927

Anderson-Darling Test for Group F:
h = 0, p = 0.050486, AD Statistic = 0.60279

Anderson-Darling Test for Group G:
h = 1, p = 0.0091747, AD Statistic = 0.79955
Warning: P is less than the smallest tabulated value, returning 0.0005.

Anderson-Darling Test for Group H:
h = 1, p = 0.0005, AD Statistic = 1.2024

```

Figure 5.32 Output for A-D test for pressure drop

Conclusion of the tests:

Since all the Anderson-Darling tests returns p values less than 0.05 (except two, that is heat transfer for 2.75 mm extrusion plate and pressure drop for 2.50 mm extrusion plate), therefore the null hypothesis of the test is rejected for all the other groups except for the two and the data can not assumed to be normally distributed.

5.4 Kruskal-Wallis anova test for verification of effect of extrusion height

Since the normality test for all groups (except two groups) fails therefore parametric one-way analysis of variance test can not be performed for these groups of data. The Kruskal-Wallis analysis of variance by ranks is a non-parametric test. It is also known as H-test. This test compares the medians of two or more independent groups when the data violates the assumptions of parametric anova. This test can be performed for data which are not normally distributed. Hence for the obtained heat transfer and pressure drop data the Kruskal-Wallis anova test can be performed to check if the variation of heat transfer and pressure drop is significant with the variation in the cylindrical extrusion height.

Kruskal-Wallis anova test for heat transfer rate and pressure drop values are performed using matlab. The test is performed among the groups of data of flat plate & cylindrical extrusion plate of 0.5 mm, 1 mm, 2 mm, 2.25 mm, 2.50 mm, 2.75 mm and 3 mm. Referring to the null hypothesis, in this case if the test is passed then it is inferred that the heat transfer rate/pressure drop does not vary significantly with the change in cylindrical extrusion height and hence is independent of it.

Null and alternate hypothesis:

Null hypothesis (H_0): The medians of all the groups are the same

Alternate hypothesis (H_a): At least one difference among the medians exists

Significance level, $\alpha = 0.05$ or 95% confidence level

Sample size, $n = 5$

Total number of groups, $c = 8$

5.4.1 Kruskal-Wallis anova test for heat transfer rate

The following code is used to perform a Kruskal-Wallis anova test for heat transfer rate data. Here, A, B, C, D, E, F, G and H are the groups of data for flat plate, cylindrical extrusion plate of 0.5 mm, 1 mm, 2 mm, 2.25 mm, 2.50 mm, 2.75 mm and 3 mm respectively.

```

untitled.mlx ×
1  A = [351.6849, 351.9504, 351.9545, 351.9504, 351.9505];
2  B = [509.2796, 493.0557, 493.0598, 493.0515, 493.0515];
3  C = [552.8255, 535.4237, 535.4279, 535.4279, 535.4237];
4  D = [597.2008, 570.4429, 570.4346, 570.4388, 570.4429];
5  E = [582.8390, 581.6612, 581.6653, 581.6570, 581.6529];
6  F = [600.3113, 600.3113, 600.3071, 600.3071, 600.3071];
7  G = [615.5523, 615.5814, 615.5399, 615.5606, 615.5399];
8  H = [583.1003, 577.8996, 577.8499, 577.8457, 577.8872];
9
10 all_data = [A, B, C, D, E, F, G, H];
11
12 group = [repmat({'A'}, 1, numel(A)), ...
13          repmat({'B'}, 1, numel(B)), ...
14          repmat({'C'}, 1, numel(C)), ...
15          repmat({'D'}, 1, numel(D)), ...
16          repmat({'E'}, 1, numel(E)), ...
17          repmat({'F'}, 1, numel(F)), ...
18          repmat({'G'}, 1, numel(G)), ...
19          repmat({'H'}, 1, numel(H))];
20
21 [p, tbl, stats] = kruskalwallis(all_data, group);

```

Figure 5.33 Matlab code for Kruskal-Wallis anova for heat transfer rate

The output of the test is:

Kruskal-Wallis ANOVA Table					
Source	SS	df	MS	Chi-sq	Prob>Chi-sq
Groups	5090	7	727.143	37.28	4.14686e-06
Error	234.5	32	7.328		
Total	5324.5	39			

Figure 5.34 Kruskal-Wallis anova table for heat transfer rate

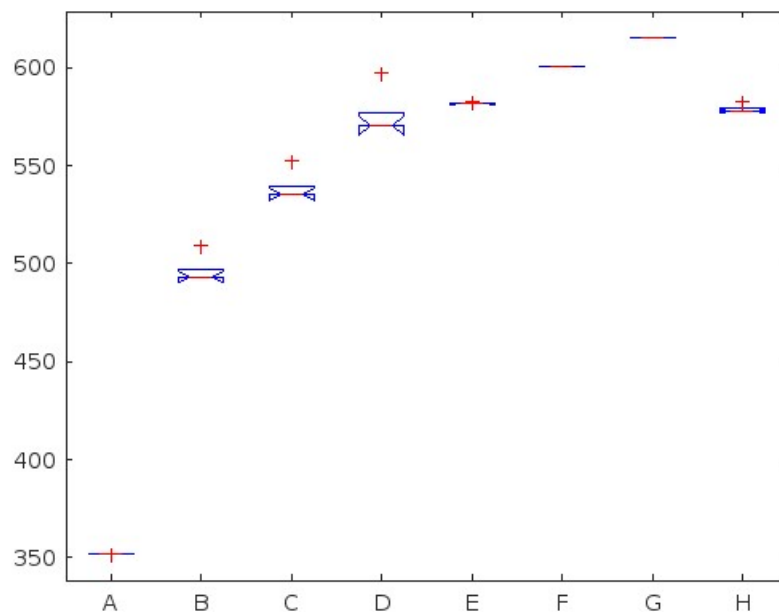


Figure 5.35 Box plot for heat transfer rate

Conclusion of the test:

The calculated p value is, $p = 4.14686 \times 10^{-6}$. Since the calculated p value is less than 0.05, therefore the null hypothesis is rejected and the alternate hypothesis is accepted (at 95% significance level). So the median of at least one of the group varies significantly from the others.

Hence it can be concluded that the variation in heat transfer rate with the variation in extrusion height is statistically significant.

5.4.2 Kruskal-Wallis anova test for pressure drop

Similarly Kruskal-Wallis anova test is performed for pressure drop data. The following matlab code is used.

```
untitled.mlx * x +
1  A = [55.5070, 53.8440, 53.8440, 53.8440, 53.8440];
2  B = [59.1760, 57.5330, 57.5330, 57.5330, 57.5330];
3  C = [61.6900, 60.7320, 60.7320, 60.7320, 60.7320];
4  D = [68.7810, 68.2790, 68.2770, 68.2780, 68.2790];
5  E = [70.3500, 70.3870, 70.3870, 70.3780, 70.3840];
6  F = [72.2840, 72.2850, 72.2830, 72.2830, 72.2830];
7  G = [73.8320, 73.8330, 73.8330, 73.8320, 73.8320];
8  H = [76.4430, 75.4850, 75.4840, 75.4850, 75.4850];
9
10 all_data = [A, B, C, D, E, F, G, H];
11
12 group = [repmat({'A'}, 1, numel(A)), ...
13          repmat({'B'}, 1, numel(B)), ...
14          repmat({'C'}, 1, numel(C)), ...
15          repmat({'D'}, 1, numel(D)), ...
16          repmat({'E'}, 1, numel(E)), ...
17          repmat({'F'}, 1, numel(F)), ...
18          repmat({'G'}, 1, numel(G)), ...
19          repmat({'H'}, 1, numel(H))];
20
21 [p, tbl, stats] = kruskalwallis(all_data, group);
```

Figure 5.36 Matlab code for Kruskal-Wallis anova for pressure drop

The output of the test is:

Kruskal-Wallis ANOVA Table					
Source	SS	df	MS	Chi-sq	Prob>Chi-sq
Groups	5250	7	750	38.58	2.35273e-06
Error	57.5	32	1.7969		
Total	5307.5	39			

Figure 5.37 Kruskal-Wallis anova table for pressure drop

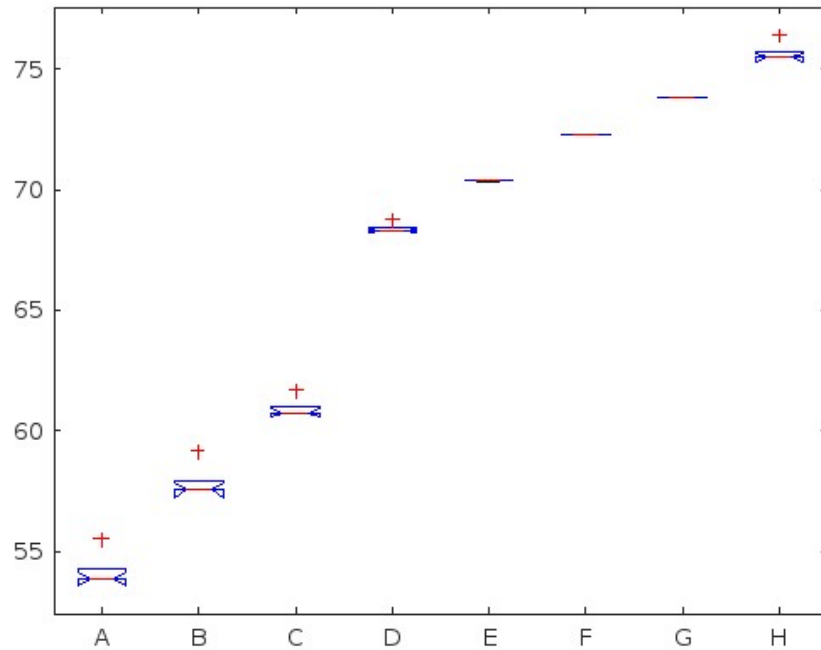


Figure 5.38 Box plot for pressure drop

Conclusion of the test:

The calculated p value is, $p = 2.35273 \times 10^{-6}$. Since the calculated p value is less than 0.05, therefore the null hypothesis is rejected and the alternate hypothesis is accepted (at 95% significance level). So the median of at least one of the group varies significantly from the others.

Hence it can be concluded that the variation in pressure drop with the variation in extrusion height is statistically significant.

5.5 Averaged heat transfer and pressure drop values

From the results of the Kruskal-Wallis anova test it is established that both heat transfer rate and pressure drop varies significantly with the variation in cylindrical extrusion height. So for further calculation the average value of heat transfer rate and pressure drop is taken.

Table 5.15 Averaged heat transfer and pressure drop values

Sl no.	Plate type	Average heat transfer rate (kW)	Average pressure drop (kPa)
1	Flat Plate	351.8989	54.6566
2	0.5 mm extrusion	496.0383	57.8616
3	1 mm extrusion	538.9058	60.9236
4	2 mm extrusion	575.7920	68.3788
5	2.25 mm extrusion	581.8951	70.3772
6	2.50 mm extrusion	600.3088	72.2836
7	2.75 mm extrusion	615.5548	73.8324
8	3 mm extrusion	578.9165	75.6824

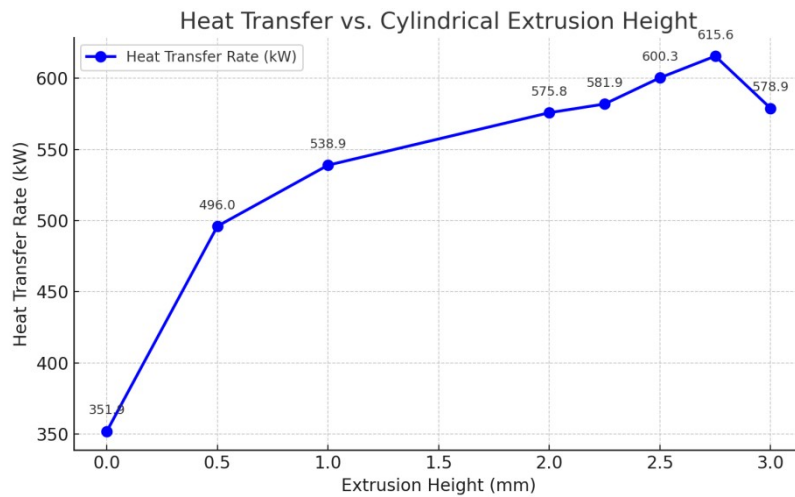


Figure 5.39 Plot for average value of heat transfer rate

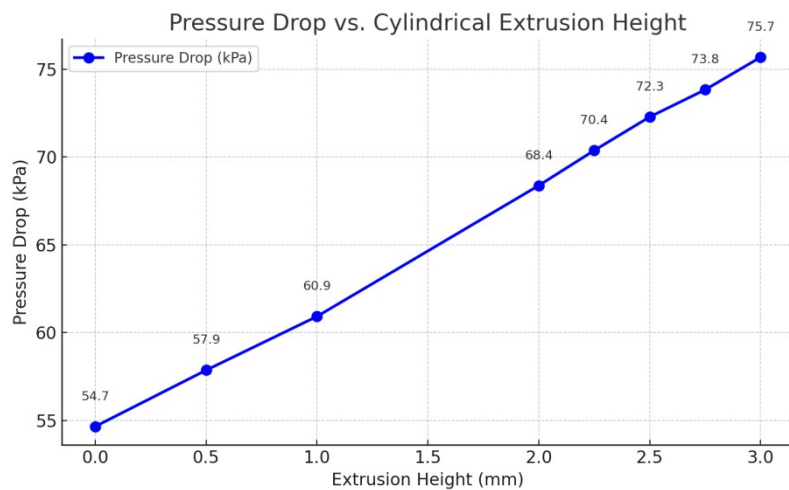


Figure 5.40 Plot for average value of pressure drop

The plot of pressure drop shows that the increase in pressure drop is almost linear and hence simple linear interpolation can be done to find the pressure drop at any point.

But the plot between average value of heat transfer rate shows the heat transfer rate keeps increasing with extrusion height in the 2 mm to 3 mm range and the drop in heat transfer rate is sudden at 3 mm.

5.6 Fitting of the curves

To proceed for optimization the above two plots are fitted using regression in matlab curve fitter app.

5.6.1 Fitting of heat transfer rate curve

The heat transfer curve is fitted to a polynomial of 5th power. The 5 degree fit has the lowest possible value of sum of squares with best goodness of fit without having signs of over fitting for the concerned range of $x = 0$ to $x = 3$ mm.

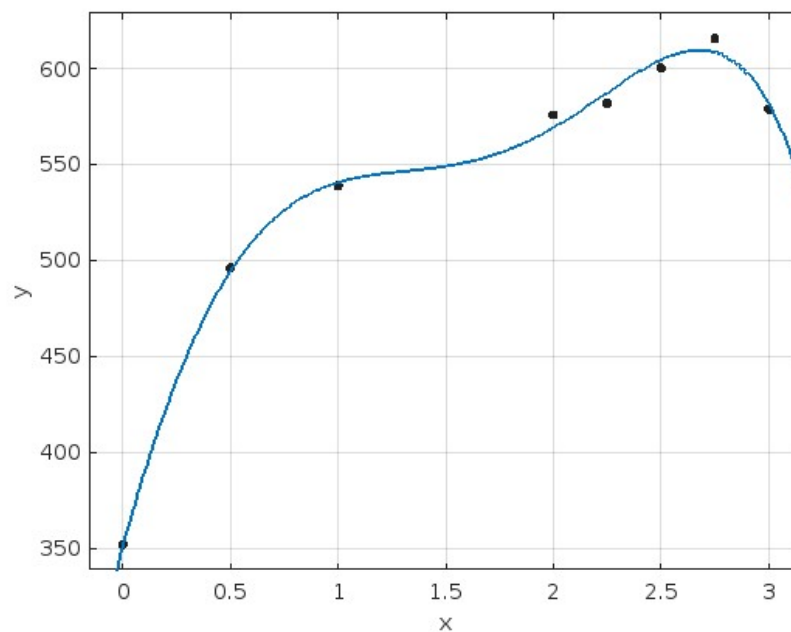


Figure 5.41 Fit plot for heat transfer rate

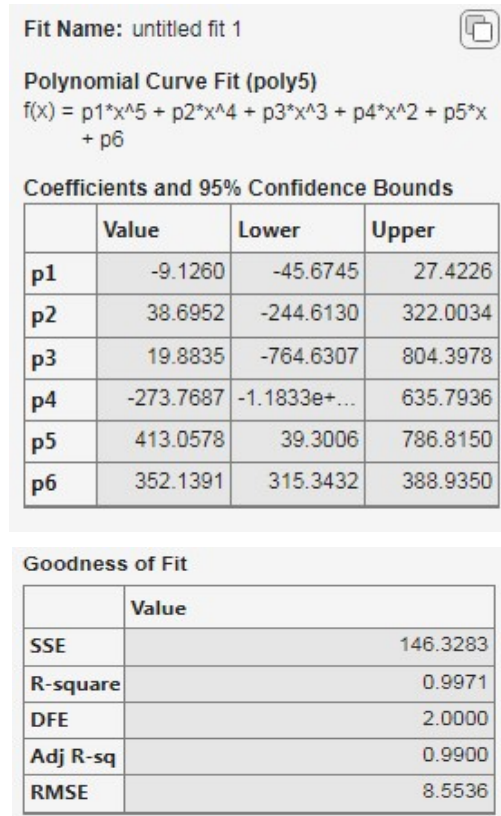


Figure 5.42 Results of fit for heat transfer rate

Therefore from the fit results the equation of variation of heat transfer with extrusion height (h_{ex}) comes to be,

$$Q = -9.126 h_{ex}^5 + 39.6952 h_{ex}^4 + 19.8835 h_{ex}^3 - 273.7687 h_{ex}^2 + 413.0578 h_{ex} + 352.1391$$

The possible explanation for the curve can be that for the region of $x = 0$ mm to almost $x = 0.5$ mm the plot follows a nearly linear curve. After $x = 2$ mm point the curve rises more sharply towards the maxima. This is possibly due to the reason that the space between the plate and the extrusion becomes very small and hence the flow between the extrusion and the plate becomes more turbulent increasing the heat transfer rate. The curve reaches maxima at almost $x = 2.75$ mm and then it fall beyond that. This may be due to the passage between the plate and the extrusion being so small that the flow is restricted in that region. Also at $x = 3$ mm the effective heat transfer area also reduces significantly due to the plate and tip of the extrusion touching each other.

5.6.2 Fitting of pressure drop curve

The pressure drop plot shows almost linear relation and hence linear fitting is used in this case. The sum of square is significantly low even in the linear fit and hence the goodness of fit is justified.

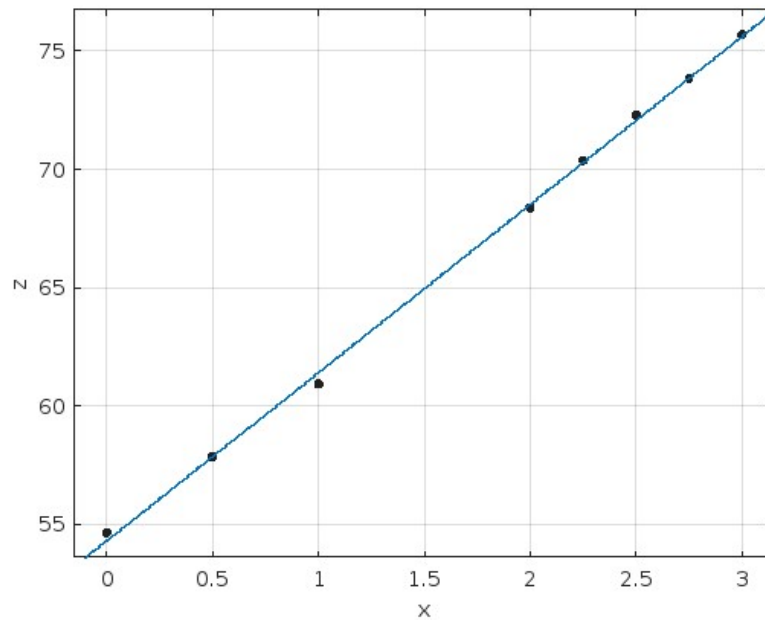


Figure 5.43 Fit plot for pressure drop

Fit Name: untitled fit 1			
Polynomial Curve Fit (poly1)			
$f(x) = p1*x + p2$			
Coefficients and 95% Confidence Bounds			
	Value	Lower	Upper
p1	7.1029	6.8783	7.3275
p2	54.3194	53.8624	54.7764
Goodness of Fit			
	Value		
SSE	0.4359		
R-square	0.9990		
DFE	6.0000		
Adj R-sq	0.9988		
RMSE	0.2695		

Figure 5.44 Results of fit for pressure drop

From the results of the linear fitting the equation for pressure drop in terms of extrusion height (h_{ex}) comes out to be,

$$\Delta P = 7.1029 h_{ex} + 54.3194$$

5.7 Optimization of the heat exchanger

To find an optimum point of extrusion height for which there is a balanced trade off between heat transfer rate and pressure drop, multi-objective optimization using normalization is done. the equations are defined as:

$$Q(h) = -9.126 h^5 + 39.6952 h^4 + 19.8835 h^3 - 273.7687 h^2 + 413.0578 h + 352.1391$$

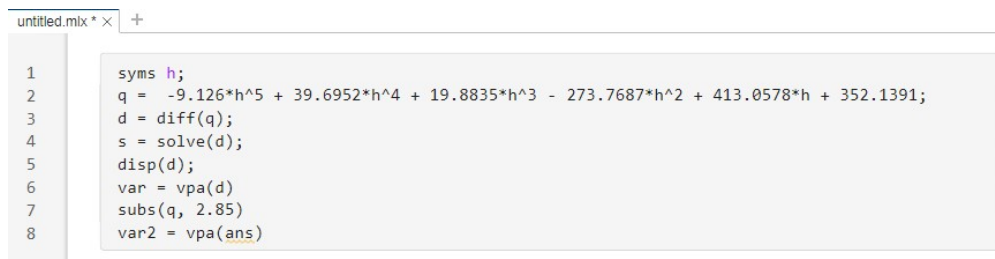
$$\text{and, } P(h) = 7.1029 h + 54.3194$$

Normalization of the equations:

The normalized equations are written as,

$$Q_n(h) = [Q(h) - Q_{\min}] / [Q_{\max} - Q_{\min}] \text{ and } P_n(h) = [P(h) - P_{\min}] / [P_{\max} - P_{\min}]$$

The maximum value of the $Q(h)$ polynomial is found using matlab,



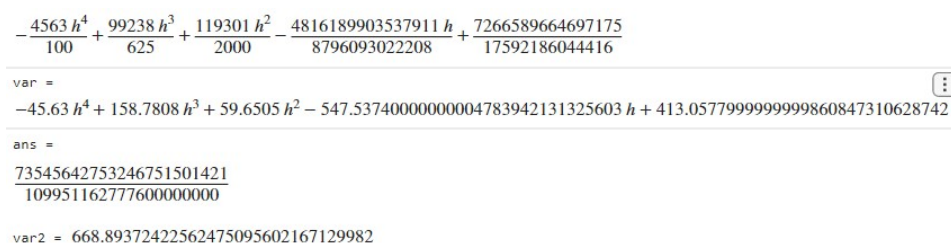
```

1  syms h;
2  q = -9.126*h^5 + 39.6952*h^4 + 19.8835*h^3 - 273.7687*h^2 + 413.0578*h + 352.1391;
3  d = diff(q);
4  s = solve(d);
5  disp(d);
6  var = vpa(d)
7  subs(q, 2.85)
8  var2 = vpa(ans)

```

Figure 5.45 Matlab code for maximizing Q

Solving the equation obtained by differentiating the equation for $Q(h)$, h is found to be 2.85 mm. So putting the value of $h = 2.85$ in the equation for $Q(h)$ we get,



$$-\frac{4563}{100} h^4 + \frac{99238}{625} h^3 + \frac{119301}{2000} h^2 - \frac{4816189903537911}{8796093022208} h + \frac{7266589664697175}{17592186044416}$$

var =

$$-45.63 h^4 + 158.7808 h^3 + 59.6505 h^2 - 547.53740000000004783942131325603 h + 413.0577999999999860847310628742$$

ans =

$$\frac{73545642753246751501421}{109951162777600000000}$$

var2 = 668.89372422562475095602167129982

Figure 5.46 Output for $Q_{\max}$

Therefore from the above results the maximum value of $Q(h)$ is obtained to be 668.8937. The minimum value for $Q(h)$ and minimum & maximum value of $P(h)$ is obtained from the plot.

$$Q_{\max} = 668.8937, \text{ at } h = 2.85$$

$$Q_{\min} = 352.1391, \text{ at } h = 0$$

$$P_{\max} = 75.6281, \text{ at } h = 3$$

$$P_{\min} = 54.3194, \text{ at } h = 0$$

Therefore the normalized equations are,

$$Q_n(h) = -0.0288 h^5 + 0.1253 h^4 + 0.0628 h^3 - 0.8643 h^2 + 1.304 h$$

$$\text{and, } P_n(h) = 0.3333 h$$

Objective function:

The composite objective function is defined as,

$$F(h) = w_1 Q_n(h) - w_2 P_n(h)$$

$$F(h) = (-0.0288 h^5 + 0.1253 h^4 + 0.0628 h^3 - 0.8643 h^2 + 1.304 h) w_1 - (0.3333 h) w_2$$

where, w_1 and w_2 are the weights that reflect the importance of heat transfer and pressure drop respectively.

If balance between heat transfer rate and pressure drop is desired then equal weights must be given to both $Q_n(h)$ and $P_n(h)$. So, $w_1 = 0.5$ and $w_2 = 0.5$.

Objective function:

$$\text{Maximize: } F_1(h) = -0.0144 h^5 + 0.0626 h^4 + 0.0314 h^3 - 0.43215 h^2 + 0.48535 h$$

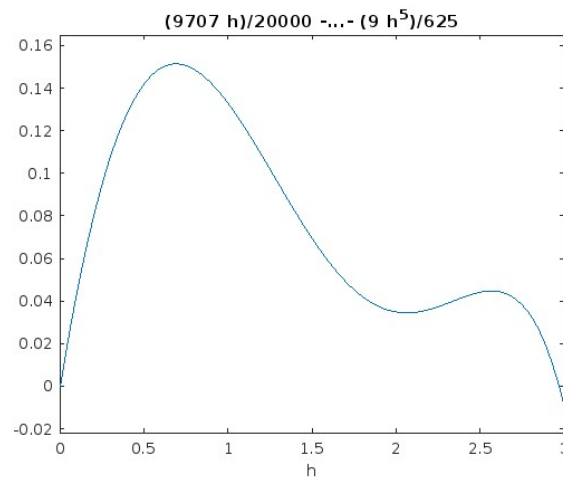


Figure 5.47 Plot of the objective function

Maximizing:

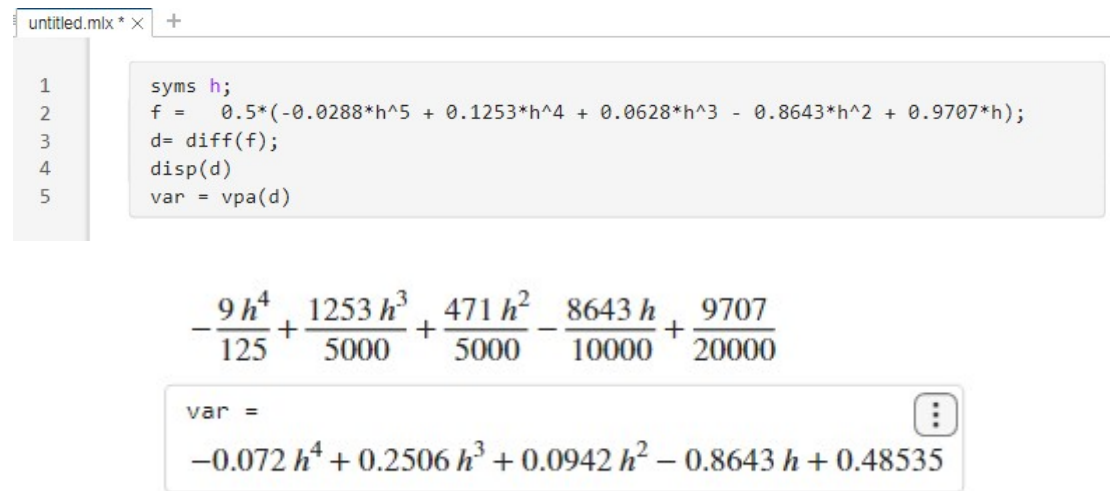


Figure 5.48 Maximization of the objective function

Solving the equation the value of extrusion height and the corresponding heat transfer rate and pressure drop is found to be,

$$h_{\text{ex}} = 0.7 \text{ mm}$$

$$Q = 521.95 \text{ kW}$$

$$\Delta P = 59.29 \text{ kPa}$$

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The results of the study showed that for a plate heat exchanger with plate with cylindrical extrusion, the heat transfer rate is enhanced considerably in comparison to flat plate design. Also this design experiences considerably less pressure drop to that of chevron plates. Anova tests showed that the extrusion height is a driving factor for the variation in the heat transfer rate and pressure drop in the heat exchanger. The pressure drop varies linearly with respect to extrusion height while the heat transfer rate plot shows multiple local maxima and minima within the range. Further by optimization of the heat exchanger it is found that for balanced trade off between heat transfer rate and pressure drop management the optimum point comes out to be 0.7 mm extrusion height. At this point the heat transfer rate is 521.95 kW, which is an 48.22% increase from the flat plate and the pressure drop is 59.29 kPa, which is an increase of 9.15% with that of the flat plate.

6.2 Future Scope

The variation of the heat transfer rate and pressure drop in the cylindrical extrusion plates with change in various parameters like extrusion spacing, size, shape etc, in addition the effect of various boundary conditions like Reynolds number, different working fluids, plate spacing is a subject of further investigation. Also repeated simulations can be performed at or around the optimum point to experimentally verify the optimization result. For the same design and working condition the optimization can also be done according to the need, for example, if the heat transfer enhancement is preferred and less priority is given to pressure drop management then more weight value must be assigned to the normalized heat transfer equation in the objective function.

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